CHAPTER 03

Bonding

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 6ce68664a9ea6d9381b821aead16f2684803b051b853969185651fe6aa18b0b1

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch03:exercise:quantum | quantum:stage-1, quantum:stage-2, quantum:stage-3, quantum:stage-4, quantum:stage-5, quantum:stage-6

ch03:exercise:bond-count | bond-count:stage-1, bond-count:stage-2, bond-count:stage-3

ch03:exercise:amide-resonance | amide-resonance:stage-1, amide-resonance:stage-2, amide-resonance:stage-3

ch03:lesson:orbitals | orbitals:rule, orbitals:cue, orbitals:trap, orbitals:app, orbitals:step-1, orbitals:step-2

ch03:lesson:sigma-pi | sigma-pi:rule, sigma-pi:cue, sigma-pi:trap, sigma-pi:app, sigma-pi:step-1, sigma-pi:step-2

ch03:lesson:hybridization | hybridization:rule, hybridization:cue, hybridization:trap, hybridization:app, hybridization:step-1, hybridization:step-2, hybridization:step-3

ch03:molecule:ethene | ethene

ch03:molecule:ethyne | ethyne

ch03:molecule:acetamide | acetamide

ch03:molecule:acetaldehyde | acetaldehyde

ch03:molecule:acetate | acetate

ch03:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 3.1 Atomic Orbitals and Quantum Numbers

scope-2: 3.2 Molecular Orbitals

scope-3: 3.3 Hybridization

Objective: Apply allowed quantum numbers

Objective: Count sigma and pi bonds

Objective: Explain hybridization and resonance

Supplied Kaplan curriculum: Chapter 3; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter03b.xhtml

OpenStax: quantum numbers, shapes and nodes | https://openstax.org/books/chemistry-atoms-first-2e/pages/3-3-development-of-quantum-theory

OpenStax: electronic structure, subshell energies and filling | https://openstax.org/books/chemistry-atoms-first-2e/pages/3-4-electronic-structure-of-atoms-electron-configurations

OpenStax: valence-bond overlap and relative bond strengths | https://openstax.org/books/chemistry/pages/8-1-valence-bond-theory

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

orbitals | Atomic orbitals and quantum numbers

Orbitals describe probability, not paths

orbitals:rule | Core Rules

n = 1,2,… sets the shell. l = 0,…,n−1 sets the subshell (s,p,d,f). m_l = −l,…,+l labels orbitals; m_s = ±1/2 labels electron spin. Each orbital holds at most two electrons with opposite spins. In the usual multielectron-atom model, subshell energies within one shell increase s < p < d < f (for subshells allowed in that shell). In the nonrelativistic hydrogen-like model, orbitals with the same n have equal energy; do not apply the multielectron splitting to that model.

Schematic summary: Orbitals describe probability, not paths

Chemistry diagram: Isolated carbon-atom spin occupancy model, not a molecular structure or an energy-scale diagram.

Shell n — 1, 2, …

Subshell l — 0 through n−1

Orbital and spin — m_l = −l … +l; m_s = ±1/2

orbitals:cue | Memory Cues

One s orbital; three p orbitals; five d orbitals; seven f orbitals. A p subshell can hold six electrons. For hydrogen-like atomic orbitals: total nodes = n−1; angular nodes = l; radial nodes = n−l−1. A 3p orbital has one angular node and one radial node.

Schematic summary: Choose the case that applies

s / p / d / f subshell — 1 / 3 / 5 / 7 orbitals

One orbital — At most two electrons with opposite spins

Hydrogen-like nodes — Total n−1; angular l; radial n−l−1

orbitals:trap | Common Mistakes / MCAT Traps

Orbital phase signs are wavefunction signs, not positive and negative electric charges. Electrons do not orbit like planets. s orbitals have spherical angular symmetry; a p orbital has two lobes separated by a nodal plane. Higher-n orbitals may also have radial nodes. A node is a zero of the wavefunction, not a place where an electron waits.

Schematic summary: Shortcut versus a better check

Avoid — Reading orbital phase signs as electric charges

Check — A probability/wavefunction model, not planetary paths

orbitals:app | MCAT Applications

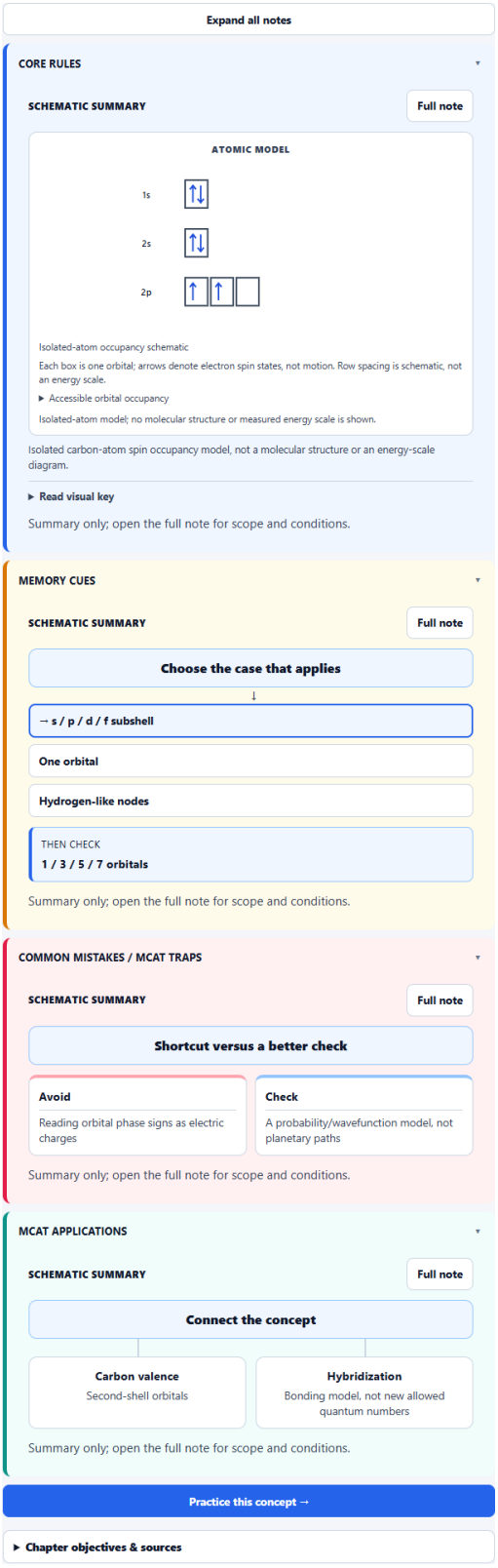
Carbon bonding uses valence orbitals from the second shell. Hybridization is a model for bonding and geometry, not a change in the allowed quantum numbers.

Schematic summary: Connect the concept

Carbon valence — Second-shell orbitals

Hybridization — Bonding model, not new allowed quantum numbers

Screenshot: orbitals-visuals



orbitals:step-1 | 1. Count valence electrons

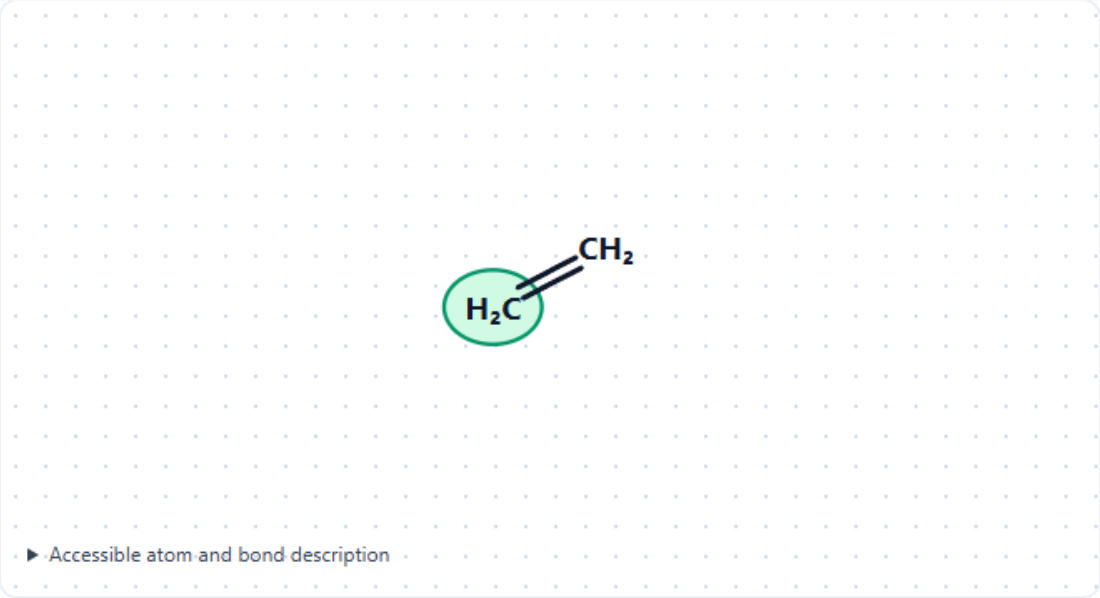
The molecular drawing is ethene, $H_2C{=}CH_2$, shown as a later bonding example. The separate box diagram describes an isolated neutral carbon atom: $1s^2 2s^2 2p^2$, with four valence electrons in the n=2 shell. Do not interpret the C=C lines as

orbital boxes or apply the isolated-atom occupancy directly to a bonded carbon.

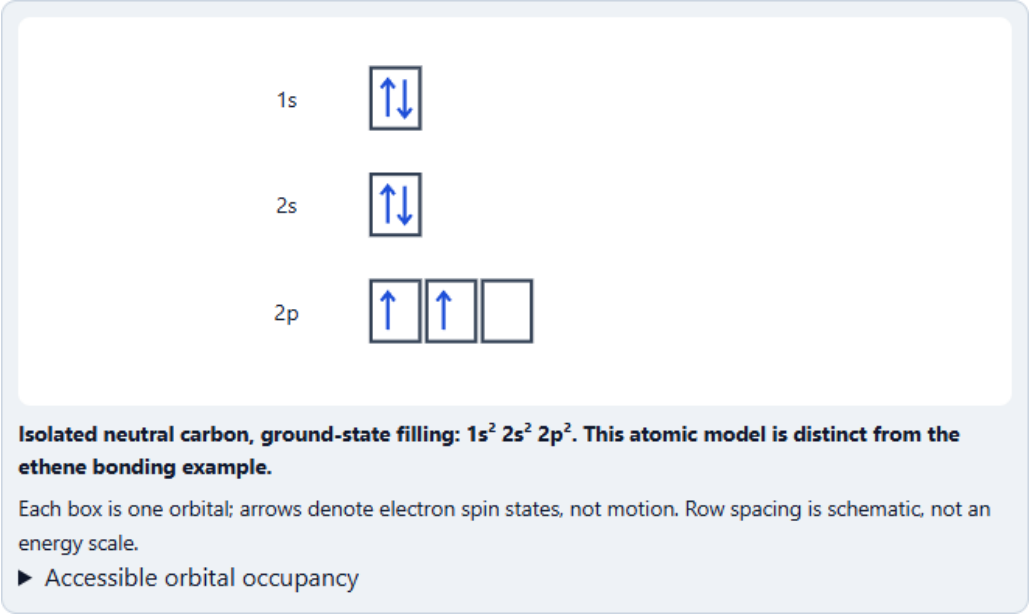
Displayed molecule ID: ethene

{"highlights": {"prin": [0]}, "revealName": true}

Screenshot: orbitals-step-1



Screenshot: orbitals-step-1-diagram



{"diagramSpecification": {"type": "orbitals", "caption": "Isolated neutral carbon, ground-state filling: 1s² 2s² 2p². This atomic model is distinct from the ethene bonding example.", "subshells": [{"label": "1s", "electrons": [2]}, {"label": "2s", "electrons": [2]}, {"label": "2p", "electrons": [1, 1, 0]}]}}

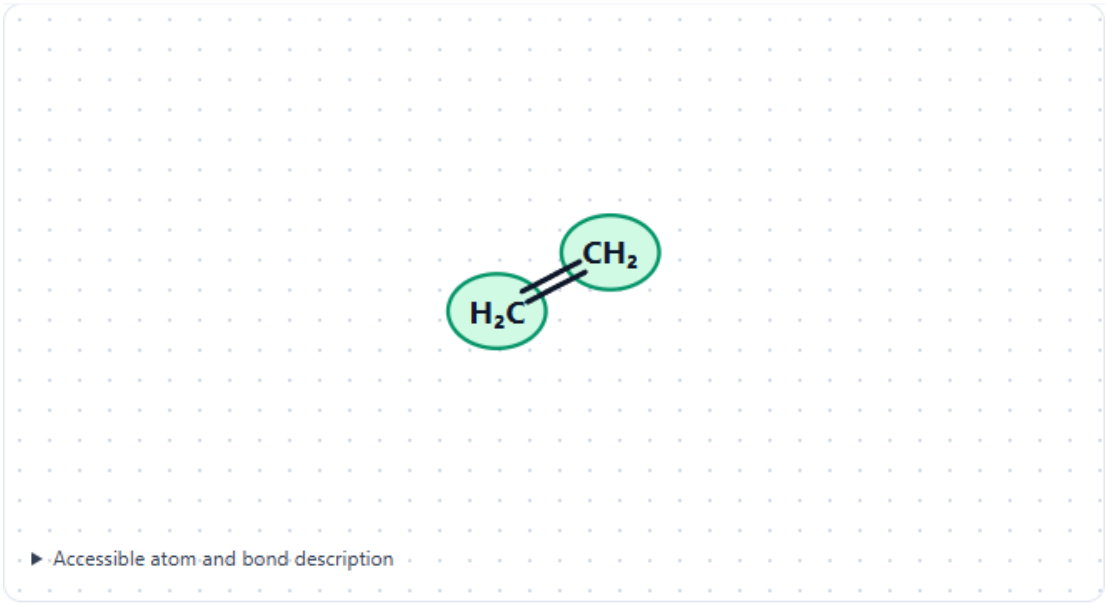
orbitals:step-2 | 2. Fill degenerate orbitals

In the isolated-carbon box diagram, the two 2p electrons occupy separate degenerate orbitals with parallel spins (Hund's rule). Electrons paired in a single orbital have opposite spins (Pauli exclusion). Ethene is the molecular example above; its bonding orbitals are a different model, developed in the next lesson.

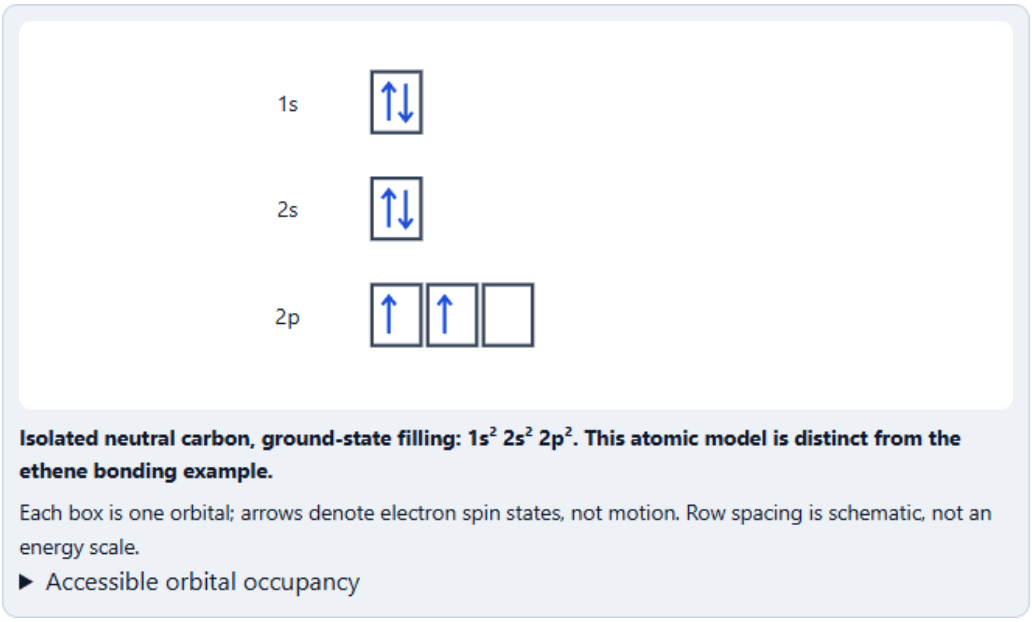
Displayed molecule ID: ethene

{"highlights": {"prin": [0, 1]}, "revealName": true}

Screenshot: orbitals-step-2



Screenshot: orbitals-step-2-diagram



{"diagramSpecification": {"type": "orbitals", "caption": "Isolated neutral carbon, ground-state filling: $1s^2\ 2s^2\ 2p^2$. This atomic model is distinct from the ethene bonding example.", "subshells": [{"label": "1s", "electrons": [2]}, {"label": "2s", "electrons": [2]}, {"label": "2p", "electrons": [1, 1, 0]}]}}

sigma-pi | Sigma, pi, and molecular orbitals

End-on vs. side-on overlap

sigma-pi:rule | Core Rules

A sigma bond has electron density along the internuclear axis. A pi bond comes from side-on p-orbital overlap. A double bond has one sigma and one pi; a triple bond has one sigma and two pi. Constructive overlap forms a lower-energy bonding MO; destructive overlap gives an antibonding MO with an added node. For comparable carbon–carbon bonds, an individual pi contribution is generally weaker than the sigma contribution because side-on overlap is less effective. Nevertheless, a double bond and a triple bond are generally stronger and shorter overall than a single bond. Their strength increments are unequal; do not add universal, context-free sigma and pi energies.

Schematic summary: End-on vs. side-on overlap

Chemistry diagram: Bonding-orbital overlap schematic. Lobe colors denote phase, not electric charge or electron paths.

Sigma — Density along the internuclear axis

Pi — Side-on p-orbital overlap

Multiple bond — Double: 1σ + 1π; triple: 1σ + 2π

sigma-pi:cue | Memory Cues

Bond order = (bonding electrons − antibonding electrons)/2 in the simple MO model. More bond order usually means shorter, stronger bonds for a given atom pair.

Schematic summary: Choose the case that applies

Simple MO bond order — Half of bonding minus antibonding electrons

Comparable atom pair — Higher bond order usually gives shorter, stronger bonds

sigma-pi:trap | Common Mistakes / MCAT Traps

A pi bond restricts rotation because twisting disrupts overlap. Bond rotation is possible only by paying an energy cost; do not imagine a double bond freely rotating at room temperature.

Schematic summary: Shortcut versus a better check

Avoid — Imagining a freely rotating double bond

Check — Twisting disrupts pi overlap and costs energy

sigma-pi:app | MCAT Applications

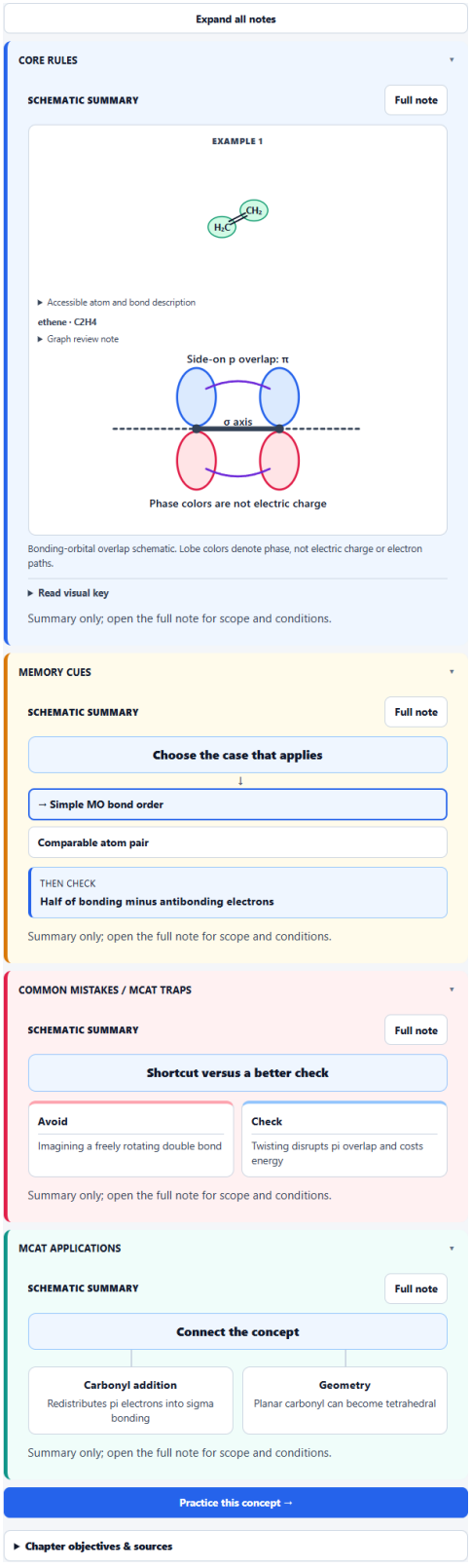
Carbonyl addition converts a pi bond into sigma bonds. Following sigma and pi electrons helps explain the change from planar to tetrahedral geometry.

Schematic summary: Connect the concept

Carbonyl addition — Redistributes pi electrons into sigma bonding

Geometry — Planar carbonyl can become tetrahedral

Screenshot: sigma-pi-visuals



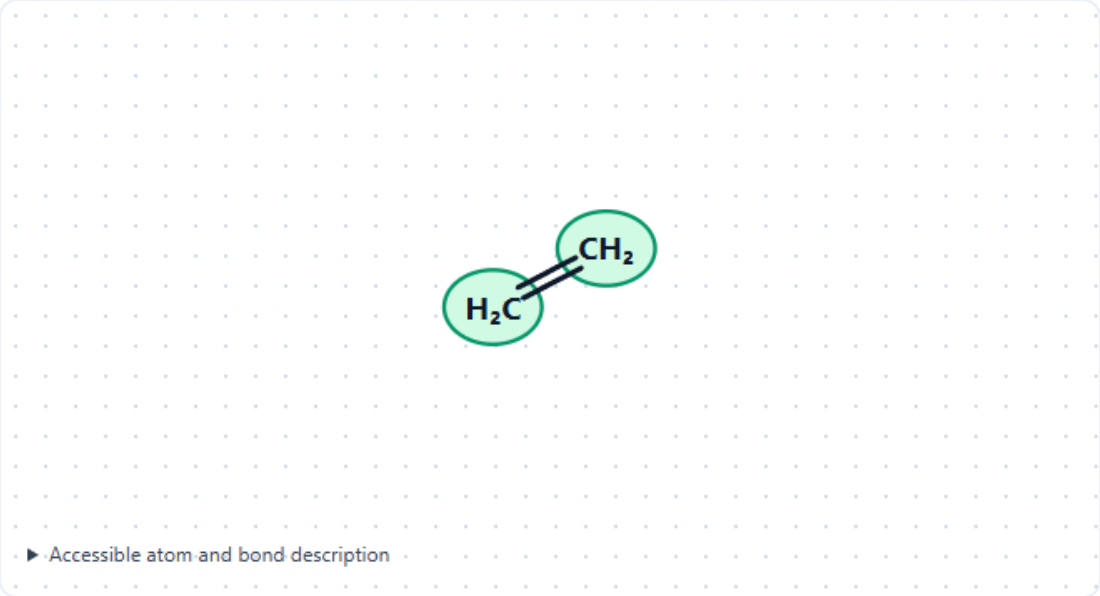
sigma-pi:step-1 | 1. Double bond inventory

Ethene has one C–C sigma bond, one C–C pi bond, and four C–H sigma bonds: five sigma and one pi overall. The H_2C and CH_2 labels show the hydrogens; the four C–H bonds are not drawn as separate lines.

Displayed molecule ID: ethene

{"highlights": {"prin": [0, 1]}, "revealName": true}

Screenshot: sigma-pi-step-1



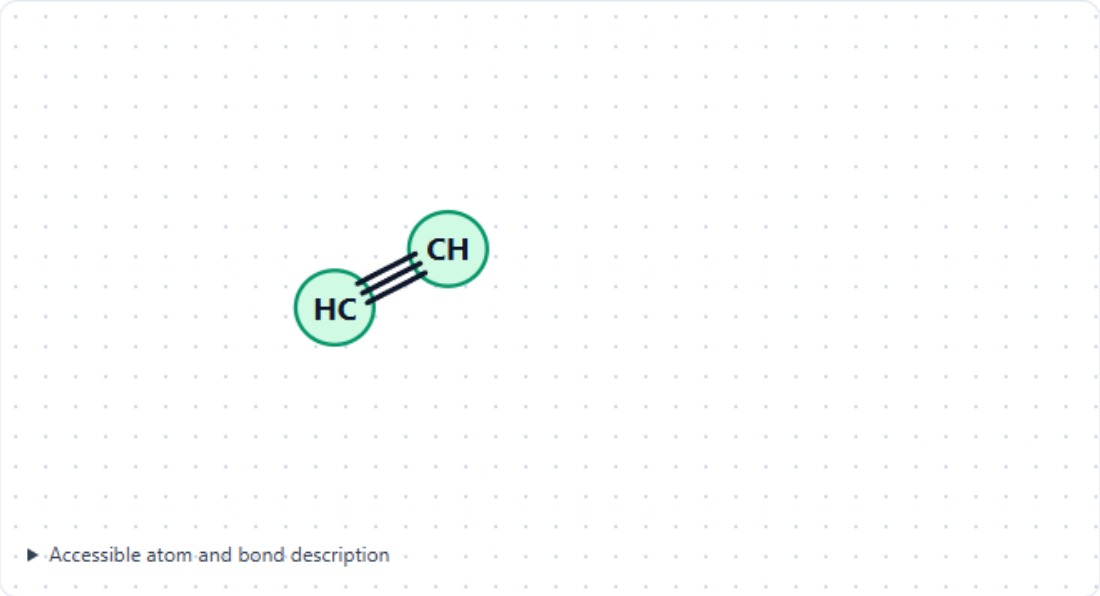
sigma-pi:step-2 | 2. Triple bond inventory

Ethyne has one C–C sigma plus two perpendicular pi bonds. Each CH label includes one hydrogen whose C–H bond is not drawn separately. Add those two C–H sigma bonds: three sigma and two pi overall.

Displayed molecule ID: ethyne

{"highlights": {"prin": [0, 1]}, "revealName": true}

Screenshot: sigma-pi-step-2



hybridization | Hybridization, geometry, and resonance

Count domains, then check conjugation

hybridization:rule | Core Rules

Typical tetrahedral carbon is sp^3 (about 109.5°), trigonal planar carbon is sp^2 (about 120°), and linear carbon is sp (180°). The s character is 25%, 33%, and 50%, respectively. Resonance contributors differ only in electron placement, not atom connectivity.

Schematic summary: Count domains, then check conjugation

Chemistry diagram: Conjugated amide and carboxylate examples. A flat drawing does not measure bond angles.

Four domains — Typical carbon: sp^3, about 109.5°

Three domains — Typical carbon: sp^2, about 120°

Two domains — Typical carbon: sp, 180°

hybridization:cue | Memory Cues

For ordinary carbon: four electron domains → sp^3; three → sp^2; two → sp. A multiple bond counts as one domain.

Schematic summary: Choose the case that applies

Ordinary carbon — A multiple bond counts as one domain

Conjugated lone pair — Check whether the simple domain shortcut applies

Amide nitrogen — Approximately planar; lone pair conjugated with C=O

hybridization:trap | Common Mistakes / MCAT Traps

Do not mechanically label an amide nitrogen sp^3 just because it has three sigma bonds and a lone pair. Its lone pair occupies a p orbital conjugated with the carbonyl, making the amide approximately planar.

Schematic summary: Shortcut versus a better check

Avoid — Assigning amide nitrogen sp^3 mechanically

Check — Conjugation and geometry, not just sigma-bond count

hybridization:app | MCAT Applications

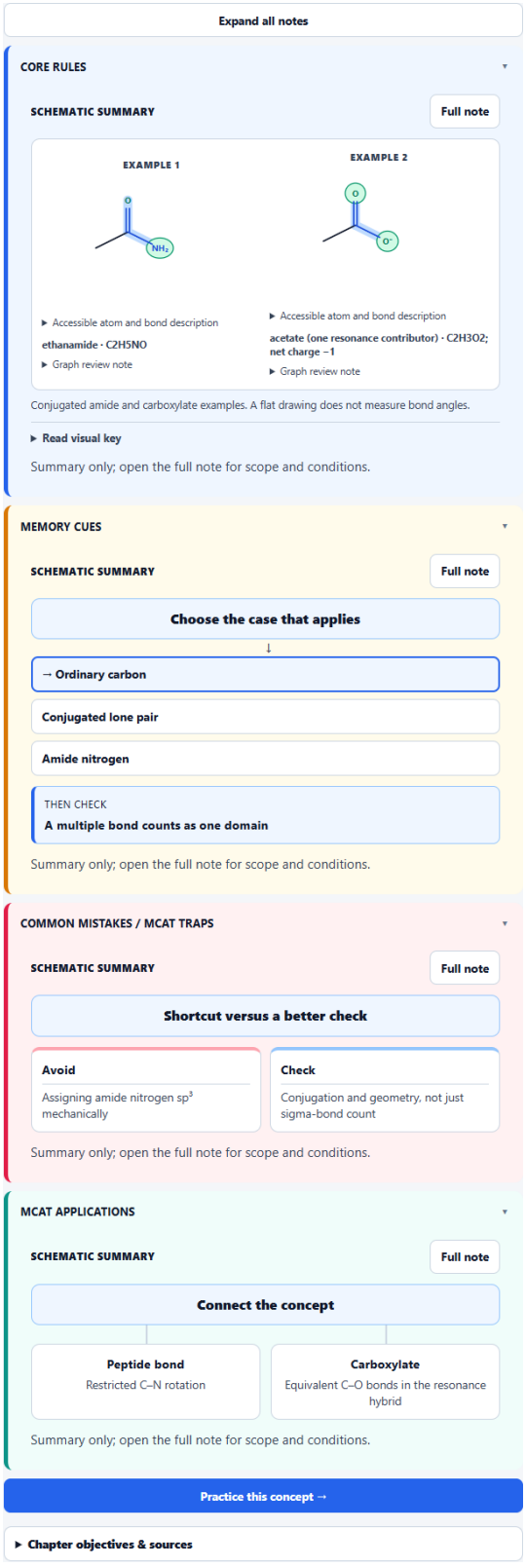
Peptide-bond resonance restricts C–N rotation. Delocalized charge explains carboxylate stability and why its two C–O bonds are equivalent in the resonance hybrid.

Schematic summary: Connect the concept

Peptide bond — Restricted C–N rotation

Carboxylate — Equivalent C–O bonds in the resonance hybrid

Screenshot: hybridization-visuals



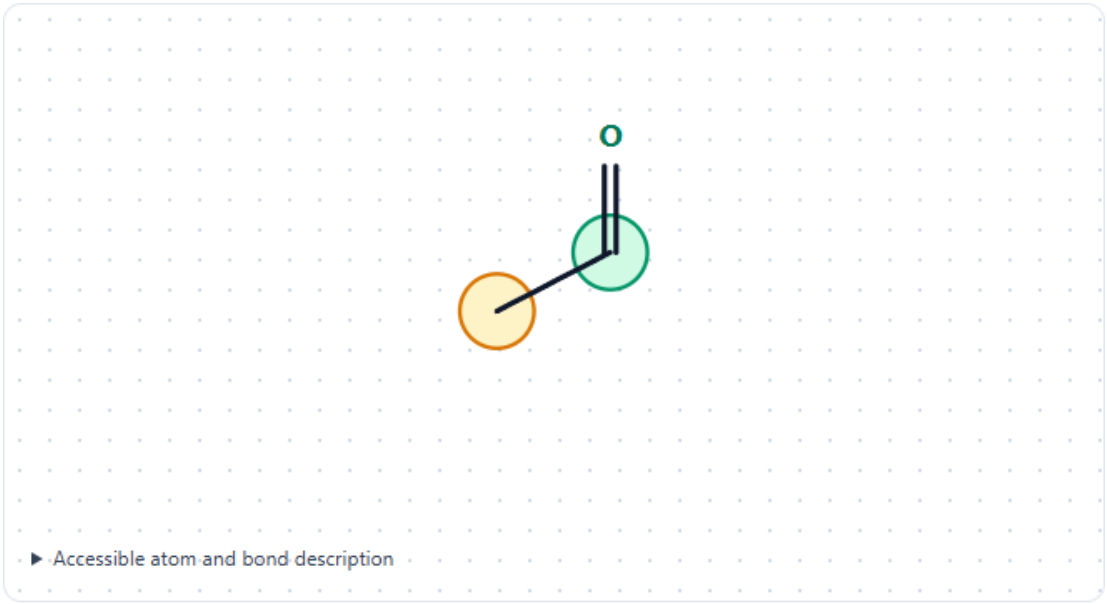
hybridization:step-1 | 1. Carbonyl geometry

The carbonyl carbon has three sigma-bond directions and a p orbital for the pi bond: sp^2 and approximately planar. Its methyl neighbor is sp^3.

Displayed molecule ID: acetaldehyde

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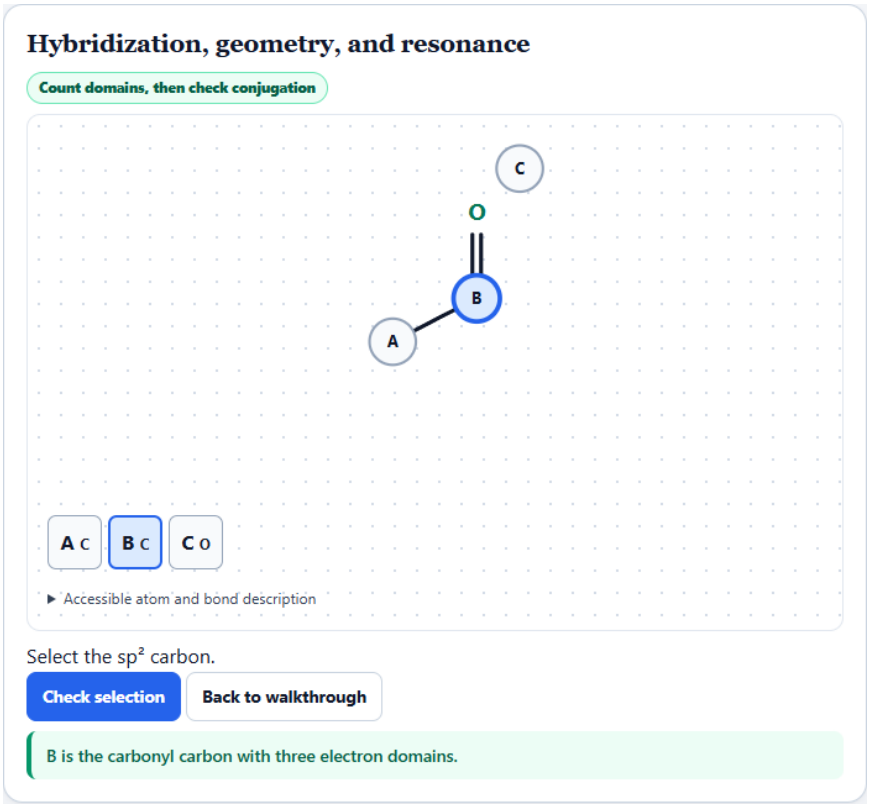
Screenshot: hybridization-step-1



Embedded challenge: Select the sp^2 carbon.

{"prompt": "Select the sp^2 carbon.", "answer": [1], "explanation": "B is the carbonyl carbon with three electron domains."}

Screenshot: hybridization-step-1-challenge



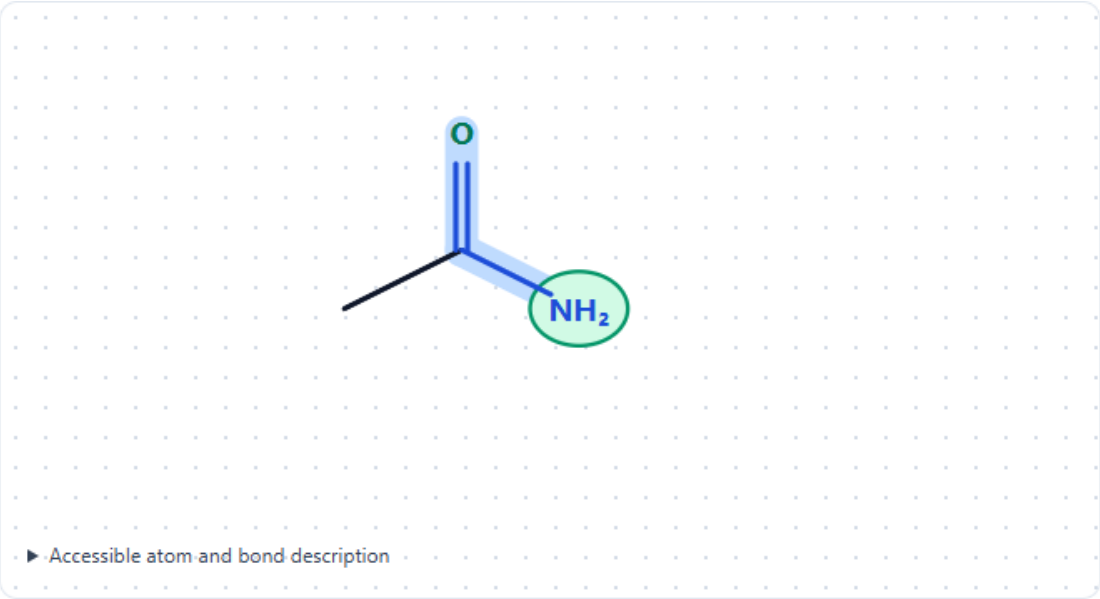
hybridization:step-2 | 2. Amide conjugation

Nitrogen donates lone-pair electron density into the carbonyl system. The C–N bond has partial double-bond character; amide nitrogen is much less basic than a typical amine.

Displayed molecule ID: acetamide

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Screenshot: hybridization-step-2



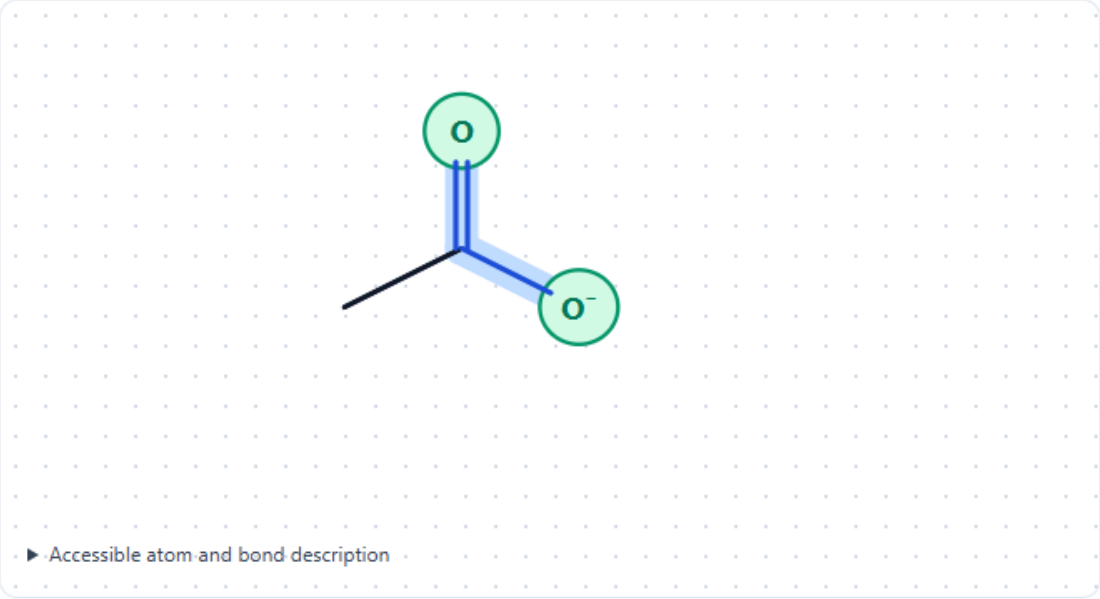
hybridization:step-3 | 3. Resonance hybrid

This drawing is one contributor of acetate. The other exchanges the two oxygen bond orders and the negative charge. The actual ion has delocalized charge; it does not flip back and forth between drawings.

Displayed molecule ID: acetate

{"highlights": {"path": [2, 1, 3], "prin": [2, 3]}, "revealName": true}

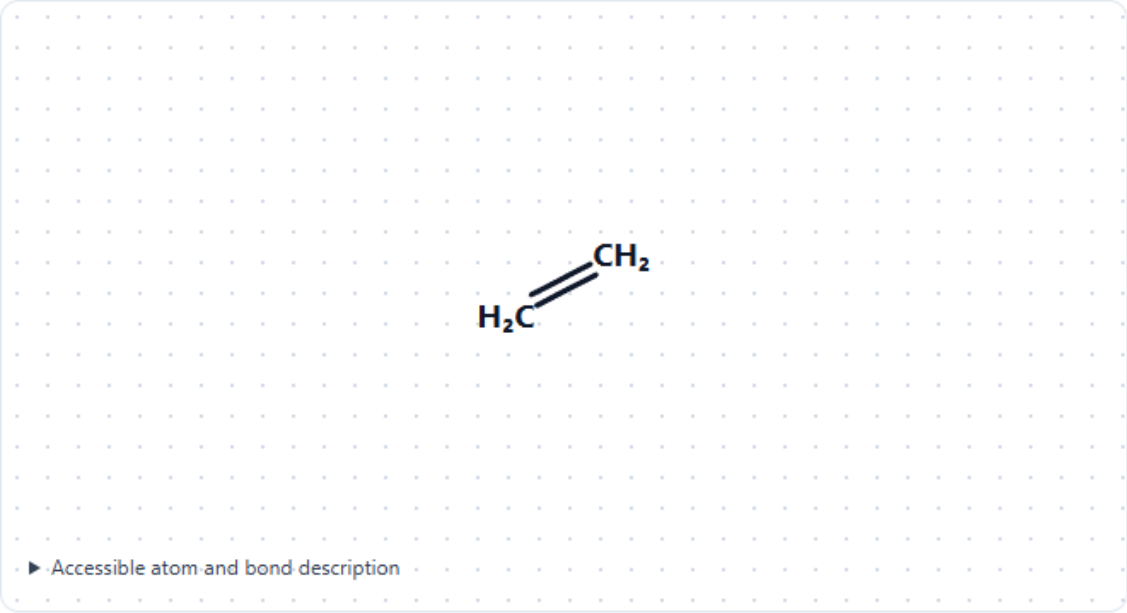
Screenshot: hybridization-step-3



quantum | Check allowed quantum numbers

The ethene drawing is a companion bonding example only. These questions concern atomic orbitals and subshells, not the total electrons or molecular orbitals of C2H4. No atomic filling diagram is supplied during this workshop.

Screenshot: quantum-stage-1-molecule



quantum:stage-1 | For a 2p electron, which n and l pair is allowed?

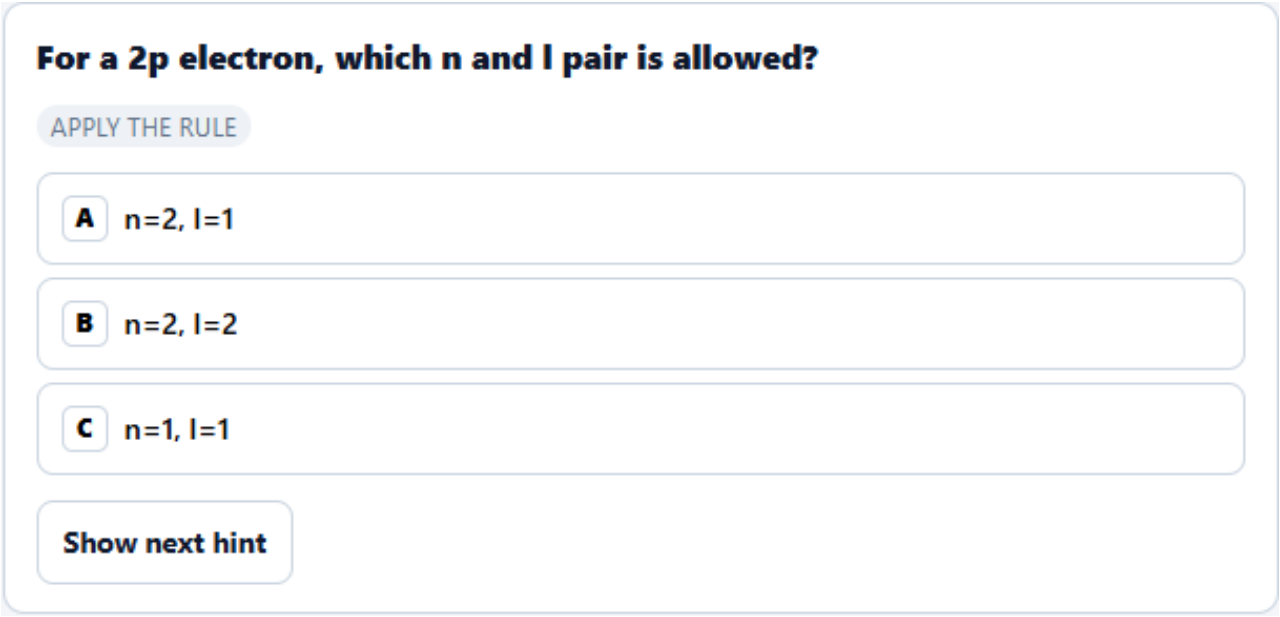
Answer type: choice

Option 0 (A): n=2, l=1

Option 1 (B): n=2, l=2

Option 2 (C): n=1, l=1

Screenshot: quantum-stage-1-question



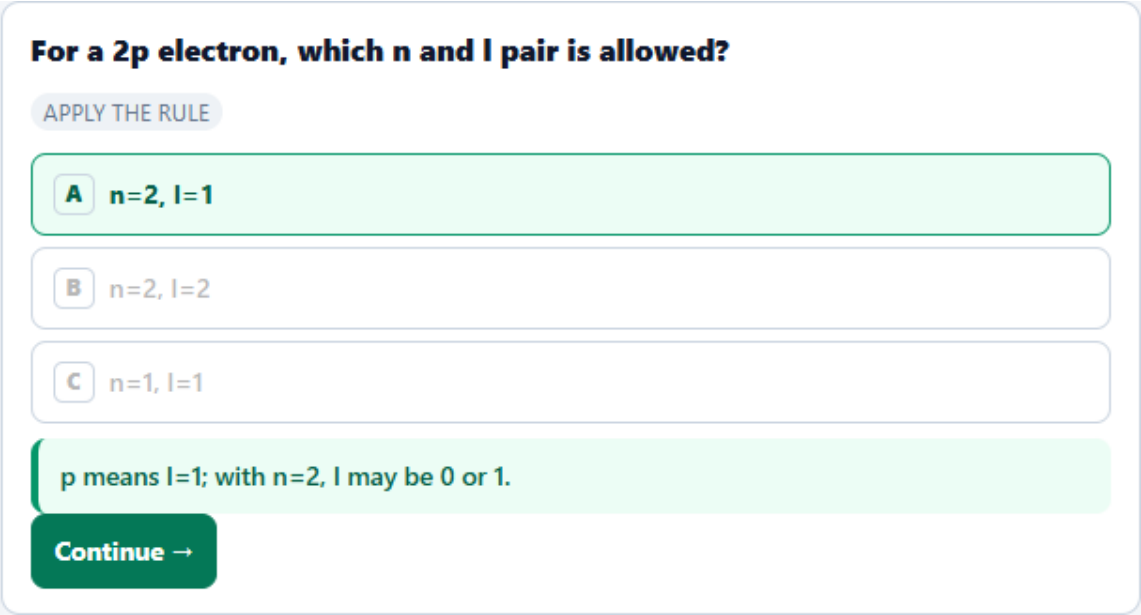
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use l ≤ n−1.

Hint 2: p is the subshell after s.

Explanation: p means l=1; with n=2, l may be 0 or 1.

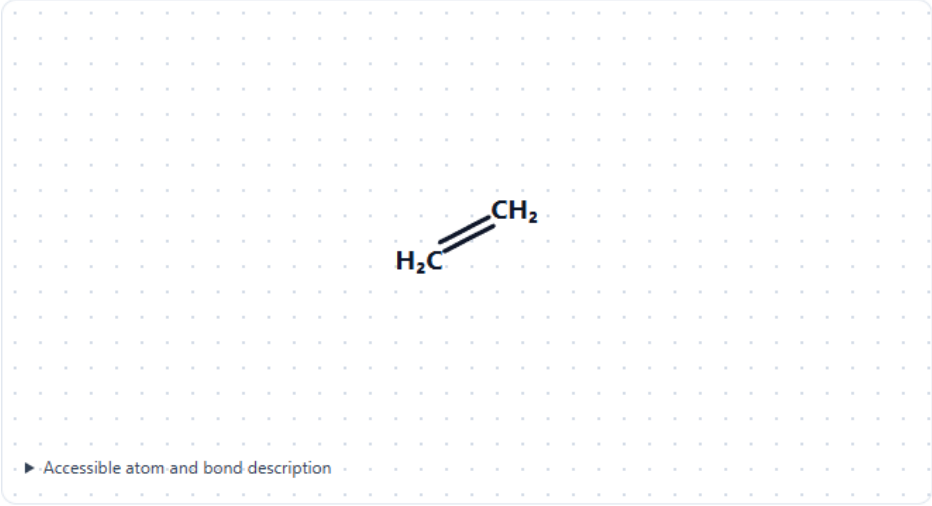
Screenshot: quantum-stage-1-feedback



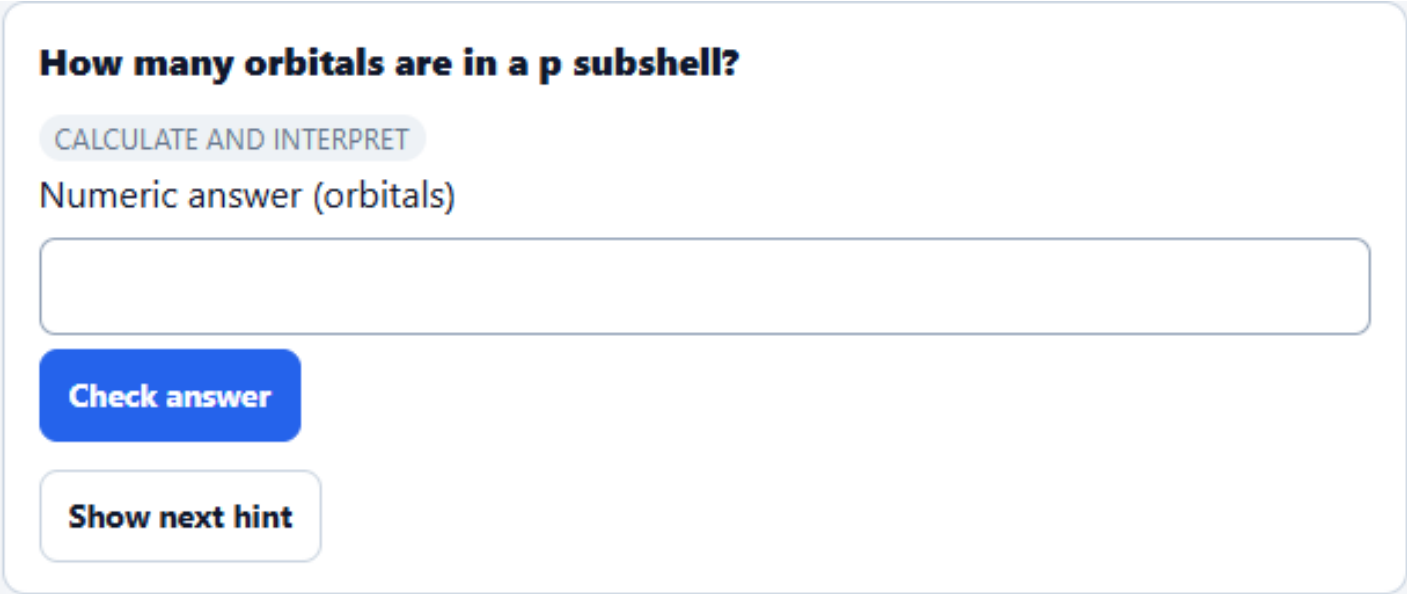
quantum:stage-2 | How many orbitals are in a p subshell?

Answer type: number | unit: orbitals

Screenshot: quantum-stage-2-molecule



Screenshot: quantum-stage-2-question



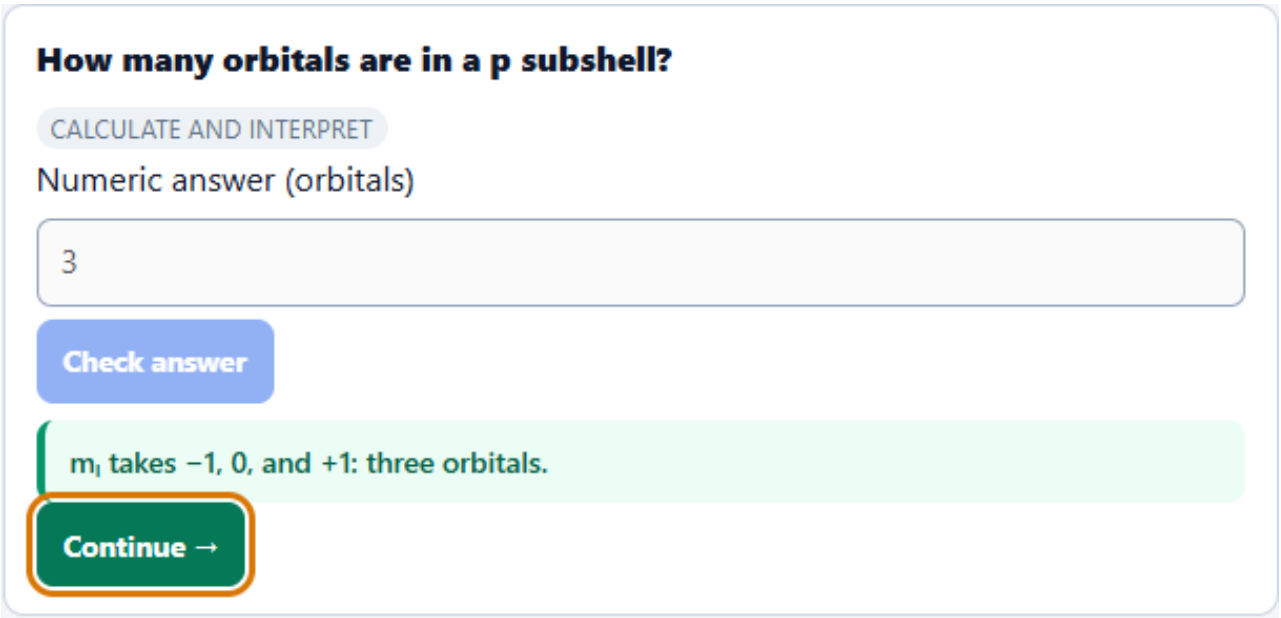
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Hint 1: Count the integers from −l to +l.

Hint 2: For p, l=1.

Explanation: m_l takes −1, 0, and +1: three orbitals.

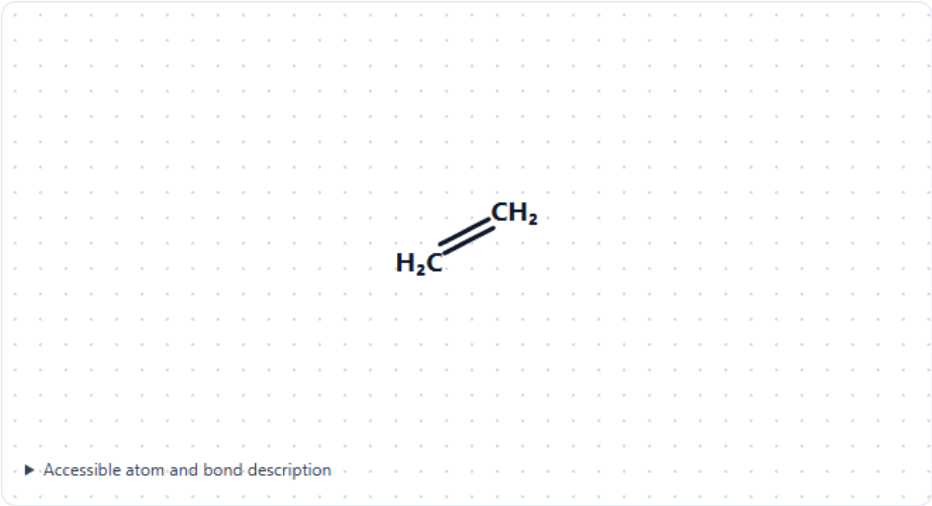
Screenshot: quantum-stage-2-feedback



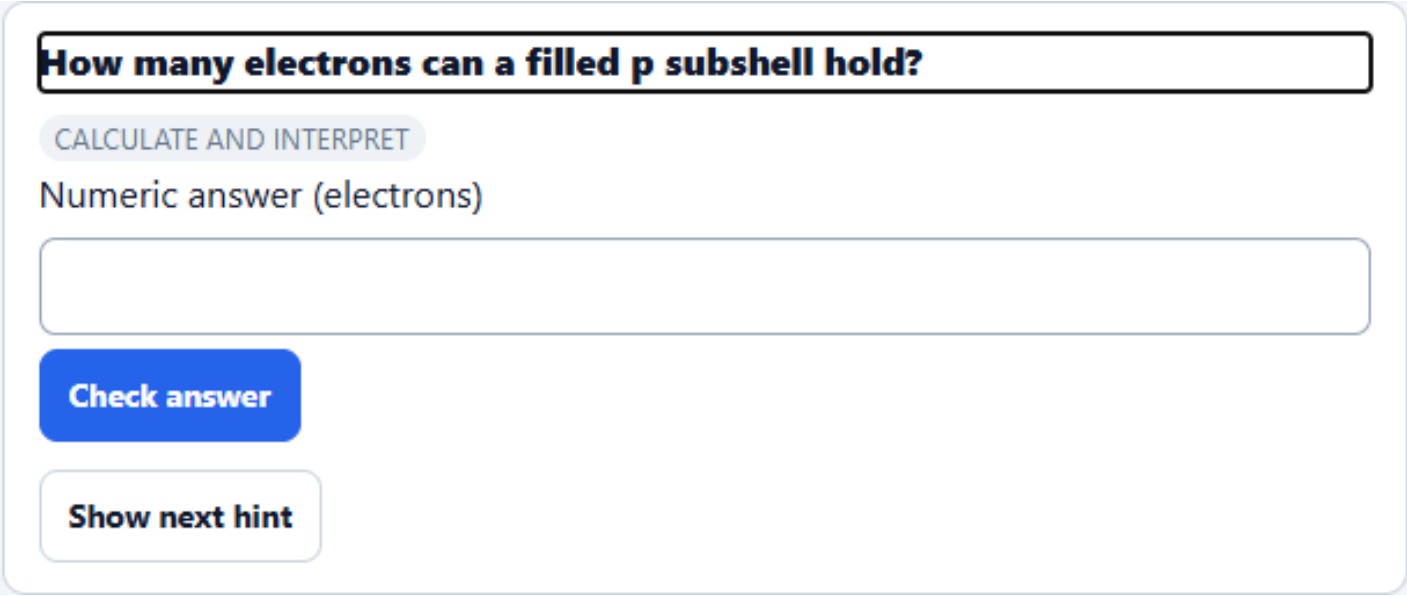
quantum:stage-3 | How many electrons can a filled p subshell hold?

Answer type: number | unit: electrons

Screenshot: quantum-stage-3-molecule



Screenshot: quantum-stage-3-question



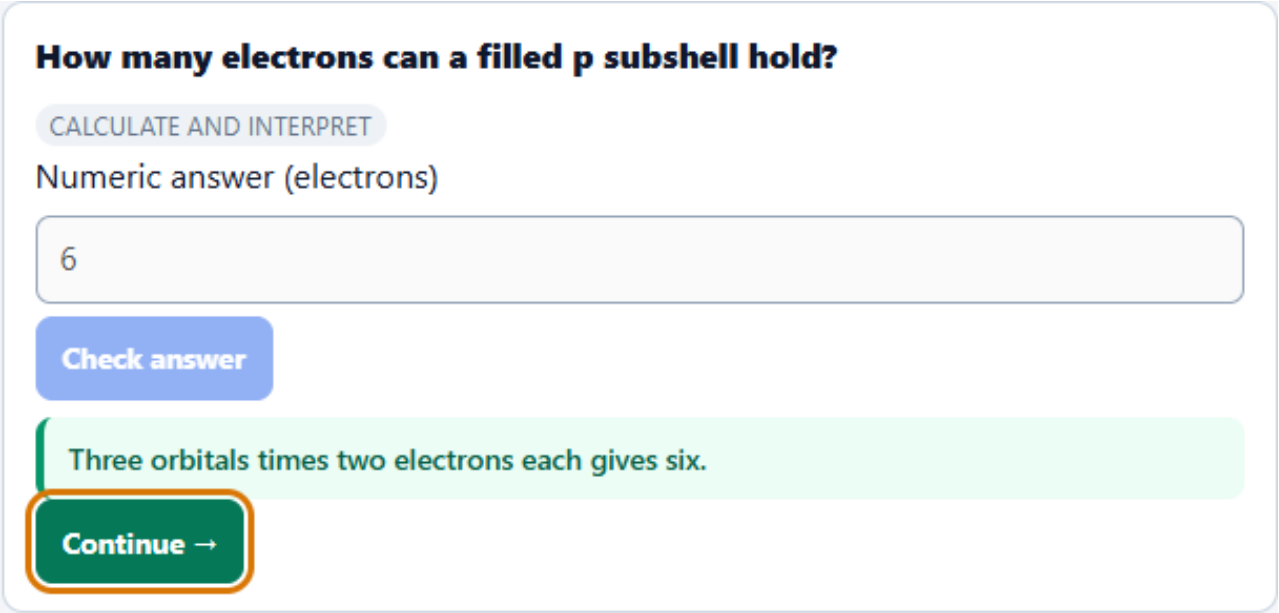
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Hint 1: Use Pauli exclusion.

Hint 2: Each orbital holds two opposite-spin electrons.

Explanation: Three orbitals times two electrons each gives six.

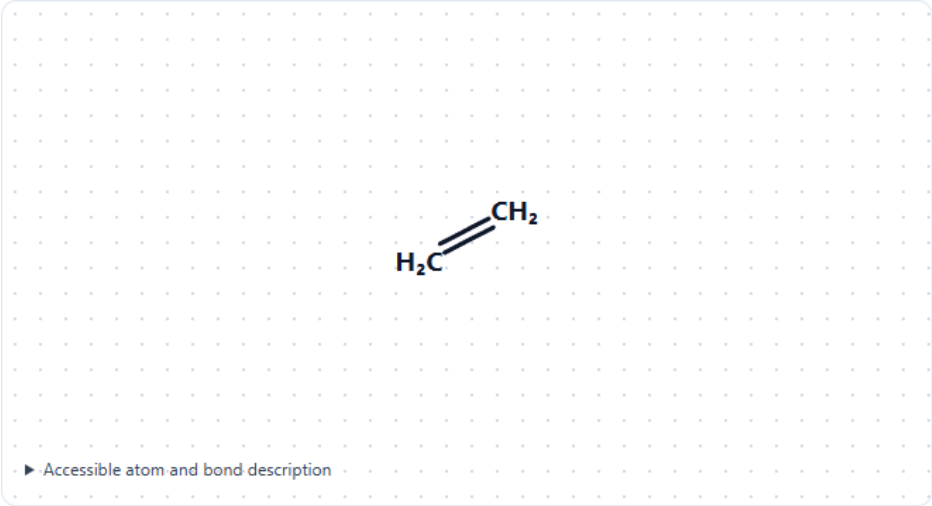
Screenshot: quantum-stage-3-feedback



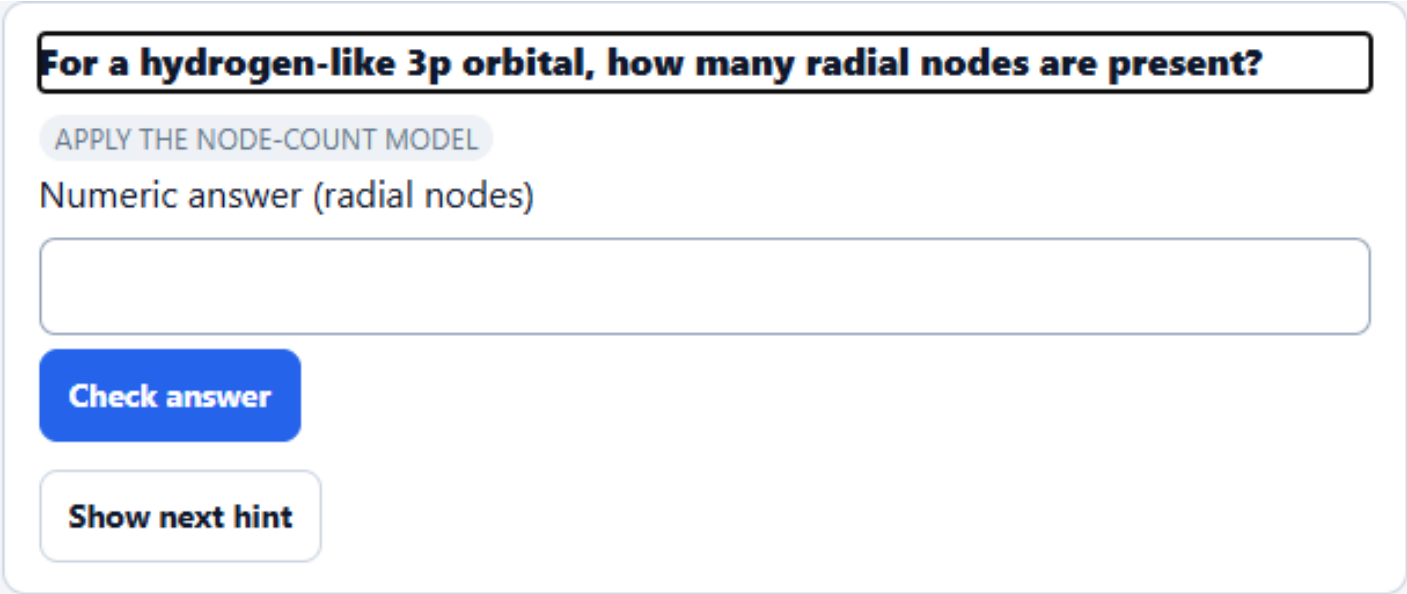
quantum:stage-4 | For a hydrogen-like 3p orbital, how many radial nodes are present?

Answer type: number | unit: radial nodes

Screenshot: quantum-stage-4-molecule



Screenshot: quantum-stage-4-question



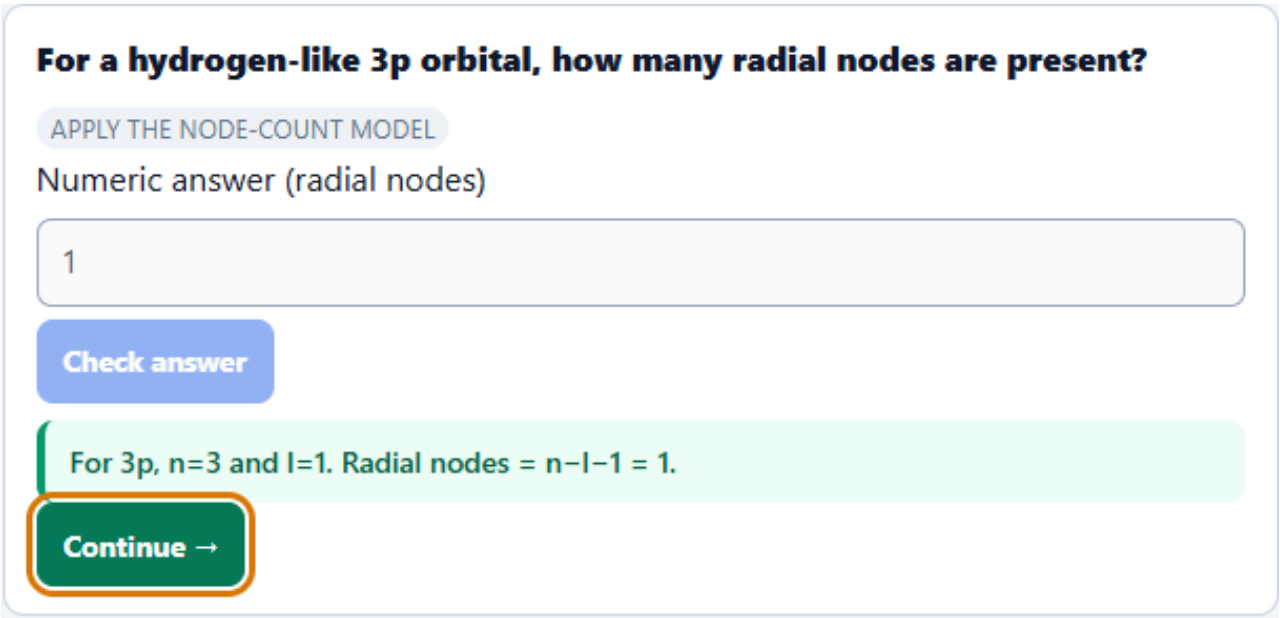
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Identify n and l from 3p.

Hint 2: Subtract l and one from n.

Explanation: For 3p, n=3 and l=1. Radial nodes = n−l−1 = 1.

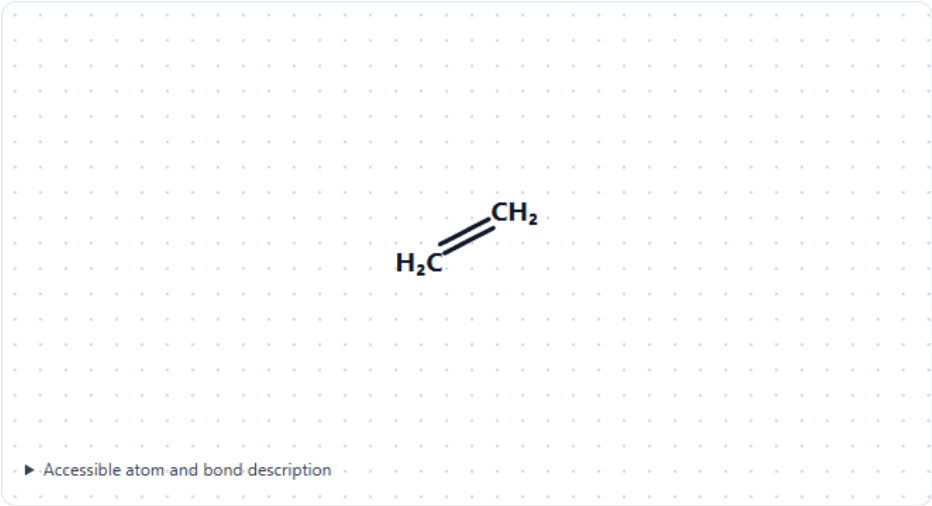
Screenshot: quantum-stage-4-feedback



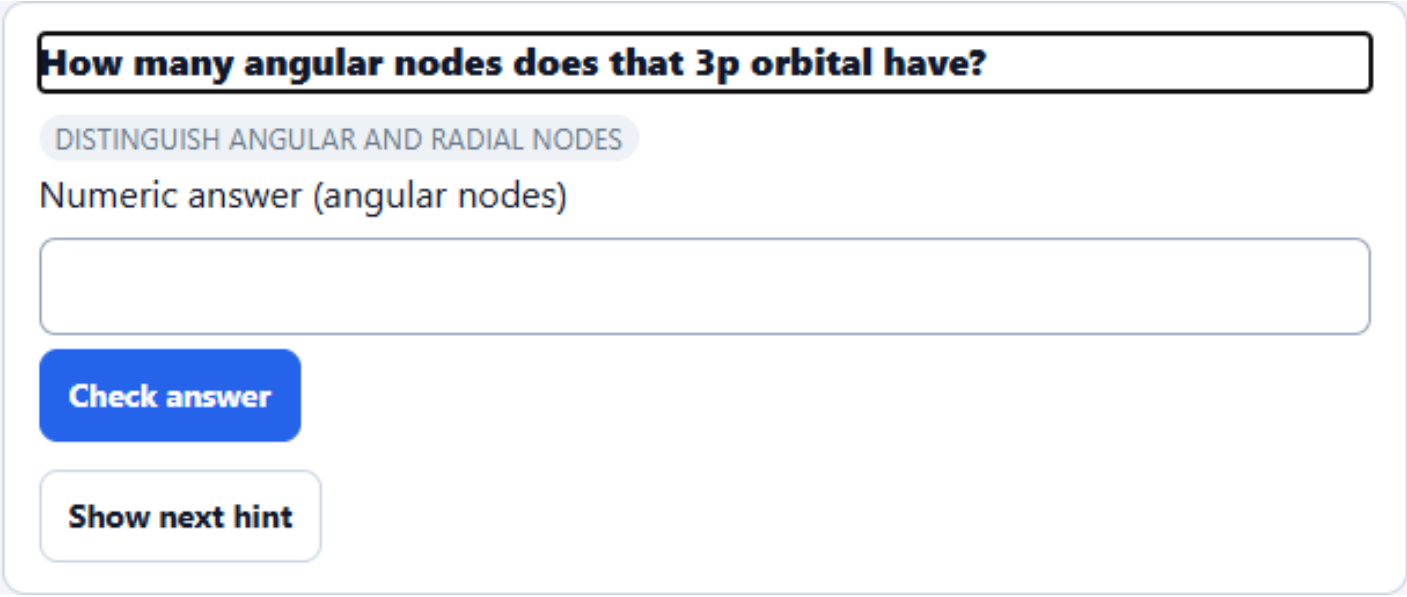
quantum:stage-5 | How many angular nodes does that 3p orbital have?

Answer type: number | unit: angular nodes

Screenshot: quantum-stage-5-molecule



Screenshot: quantum-stage-5-question



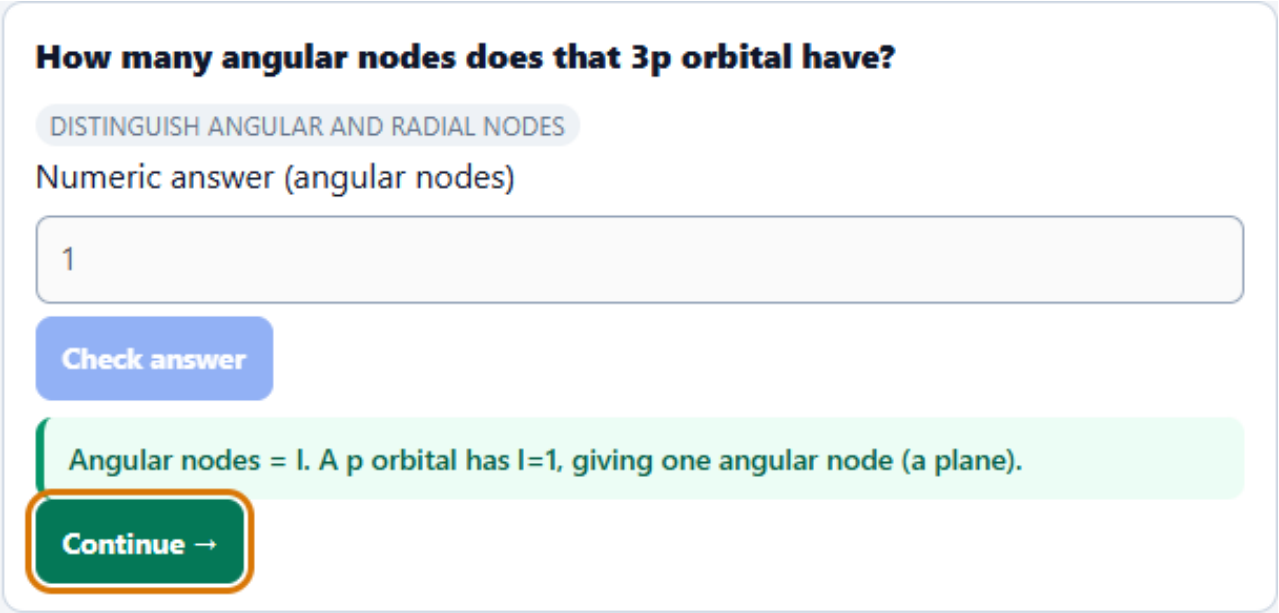
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the subshell letter rather than n.

Hint 2: The angular-node count equals l.

Explanation: Angular nodes = l. A p orbital has l=1, giving one angular node (a plane).

Screenshot: quantum-stage-5-feedback



quantum:stage-6 | In the usual multielectron-atom model, which listed n=3 subshell is lowest in energy?

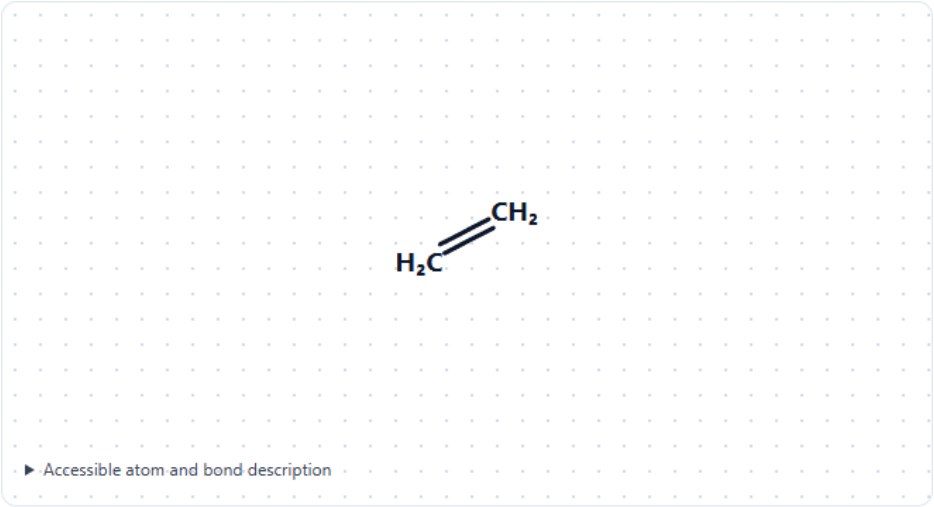
Answer type: choice

Option 0 (A): 3s

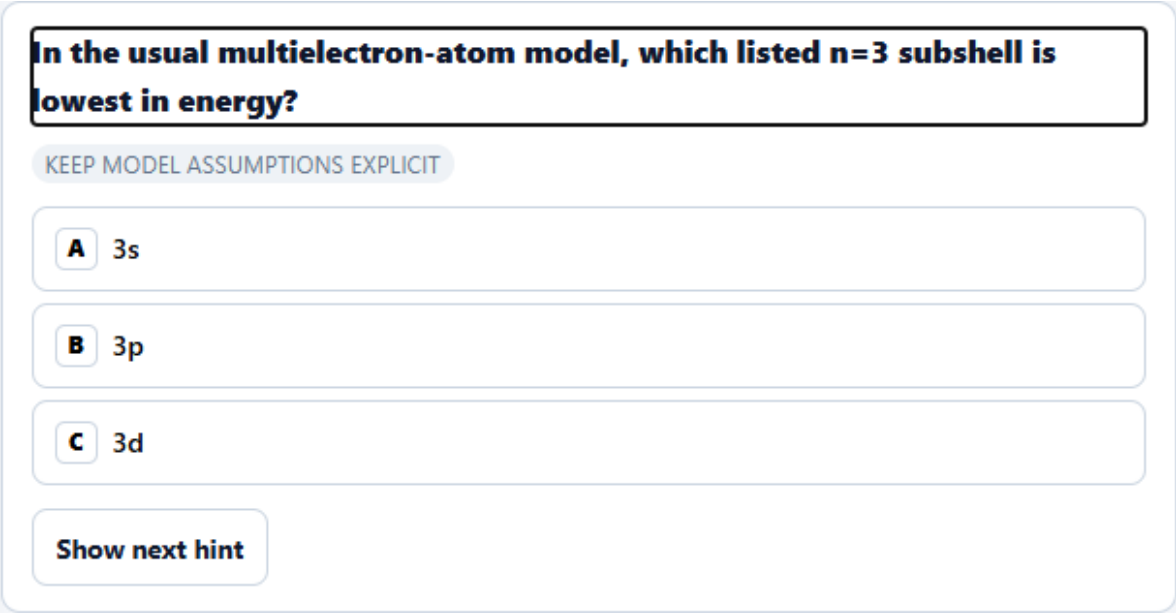
Option 1 (B): 3p

Option 2 (C): 3d

Screenshot: quantum-stage-6-molecule



Screenshot: quantum-stage-6-question



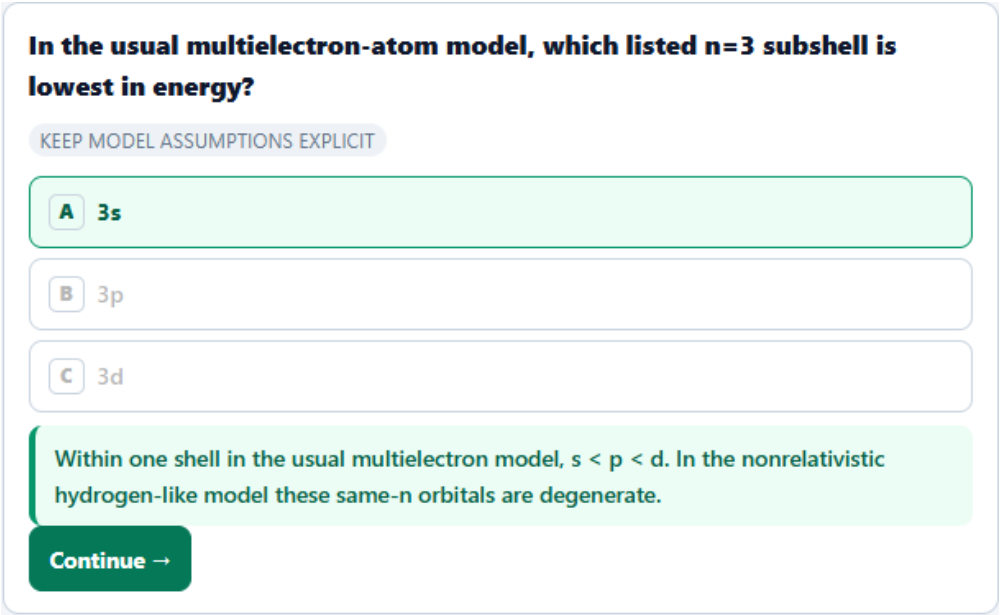
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Electron–electron interactions distinguish subshell energies.

Hint 2: Use the within-shell order s < p < d.

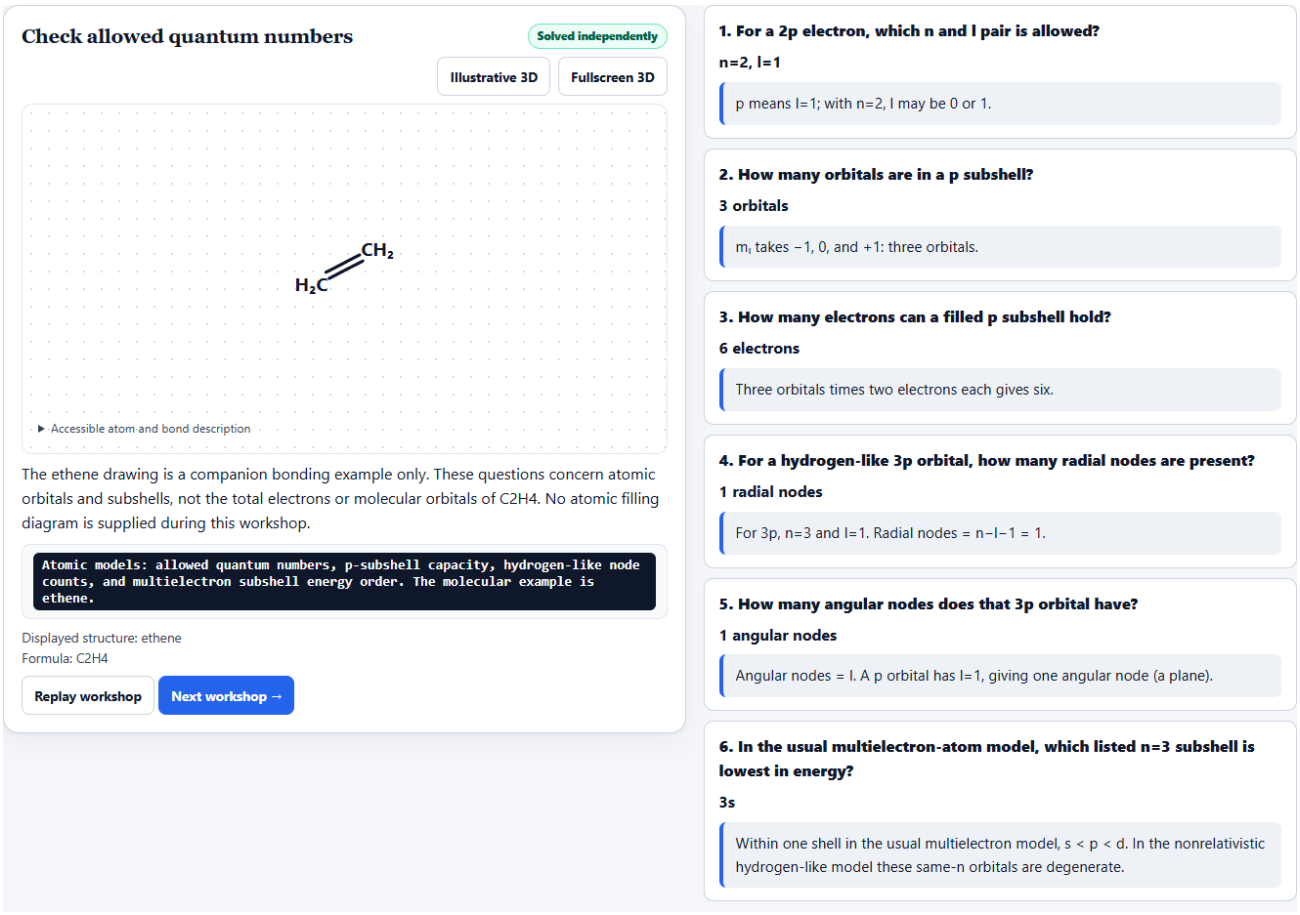
Explanation: Within one shell in the usual multielectron model, s < p < d. In the nonrelativistic hydrogen-like model these same-n orbitals are degenerate.

Screenshot: quantum-stage-6-feedback



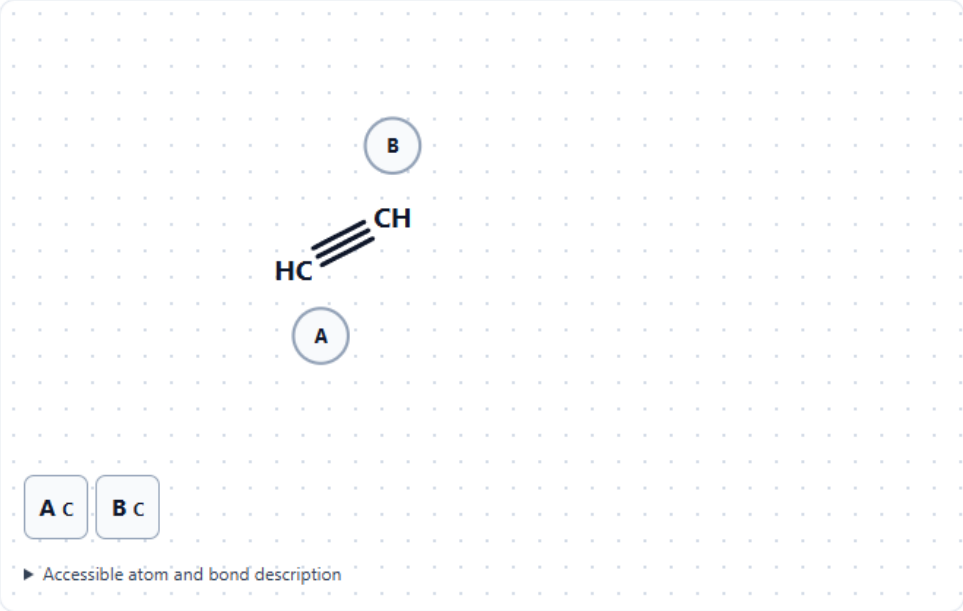
Completion summary: Atomic models: allowed quantum numbers, p-subshell capacity, hydrogen-like node counts, and multielectron subshell energy order. The molecular example is ethene.

Screenshot: quantum-complete



bond-count | Count all sigma and pi bonds

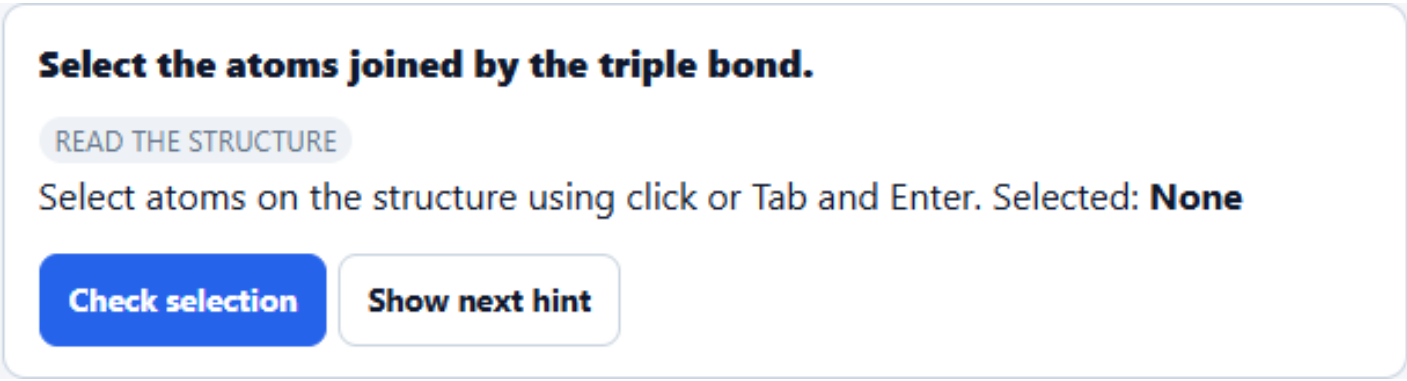
Screenshot: bond-count-stage-1-molecule



bond-count:stage-1 | Select the atoms joined by the triple bond.

Answer type: atoms

Screenshot: bond-count-stage-1-question



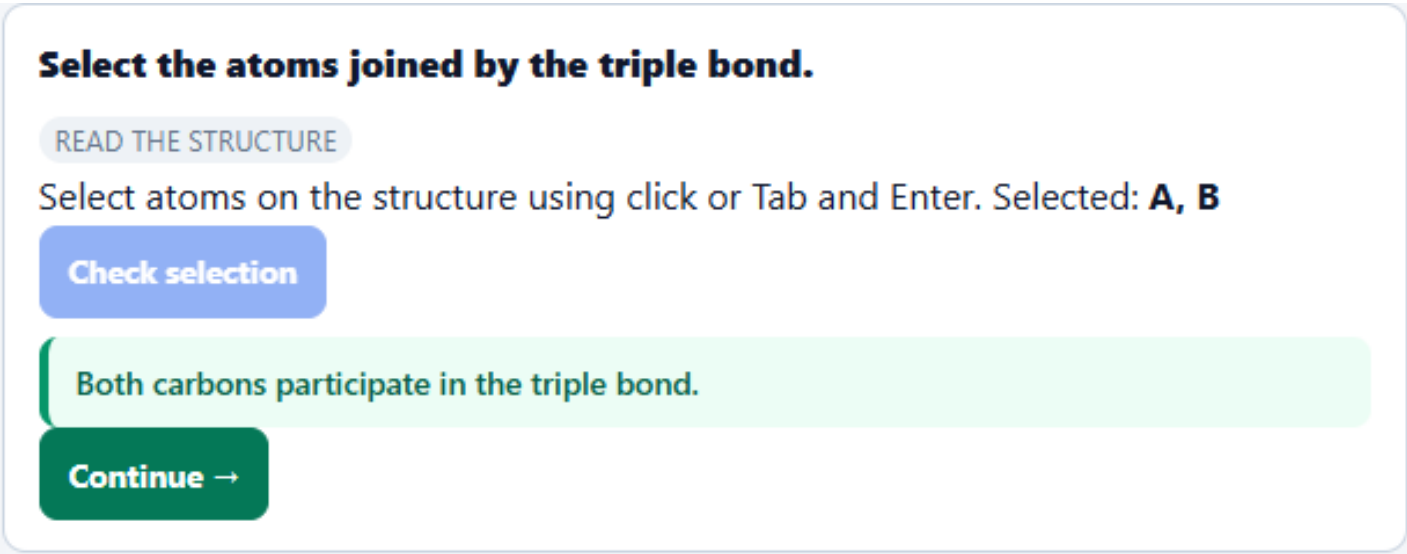
{"authorAnswer": [0, 1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 1]}}

Hint 1: Find the three parallel lines.

Hint 2: Select both ends.

Explanation: Both carbons participate in the triple bond.

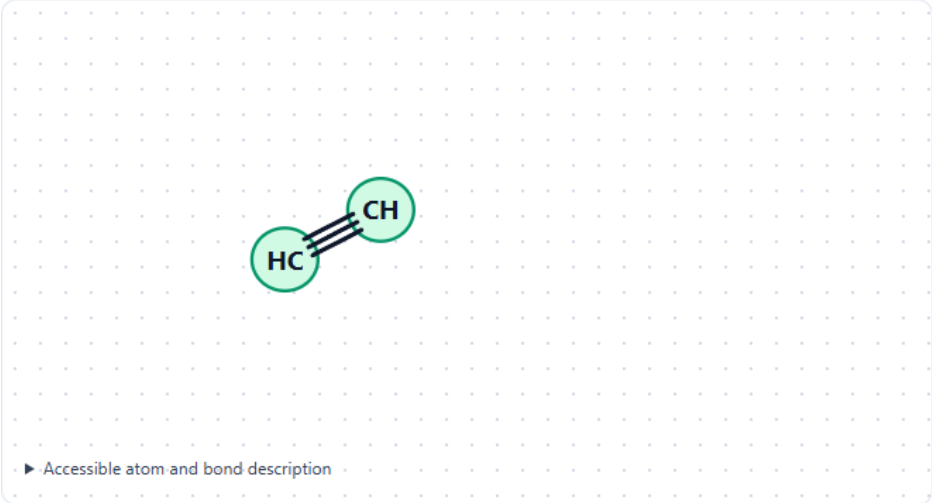
Screenshot: bond-count-stage-1-feedback



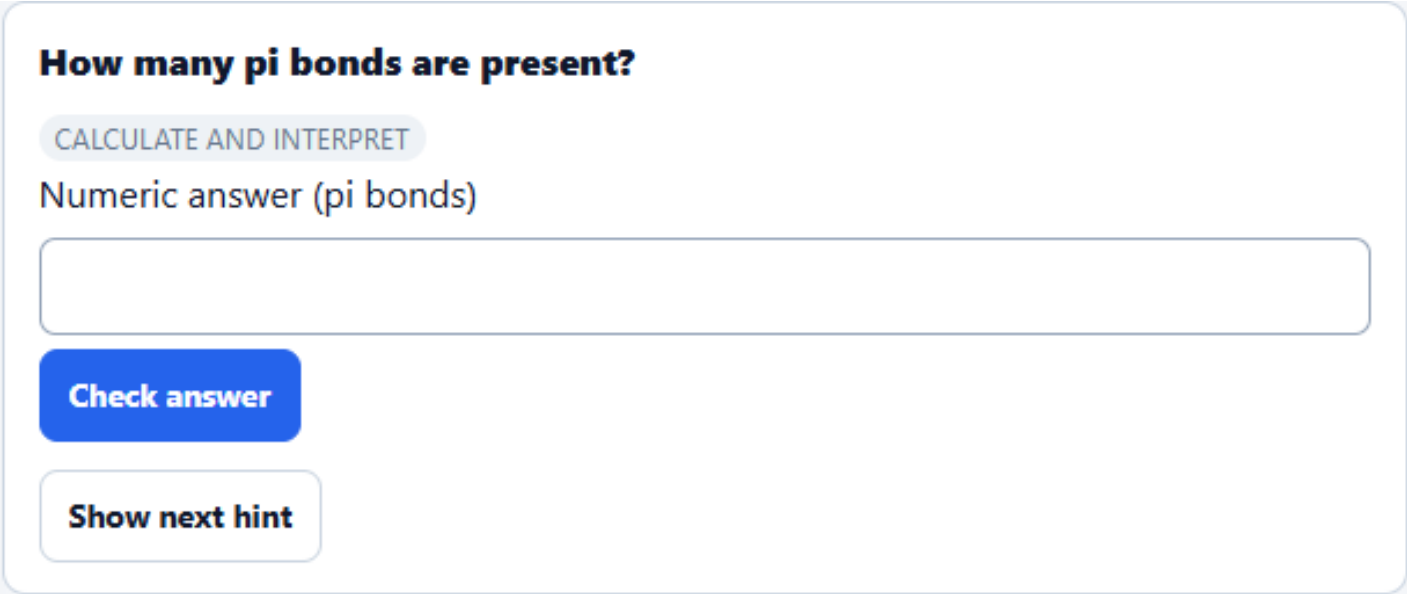
bond-count:stage-2 | How many pi bonds are present?

Answer type: number | unit: pi bonds

Screenshot: bond-count-stage-2-molecule



Screenshot: bond-count-stage-2-question



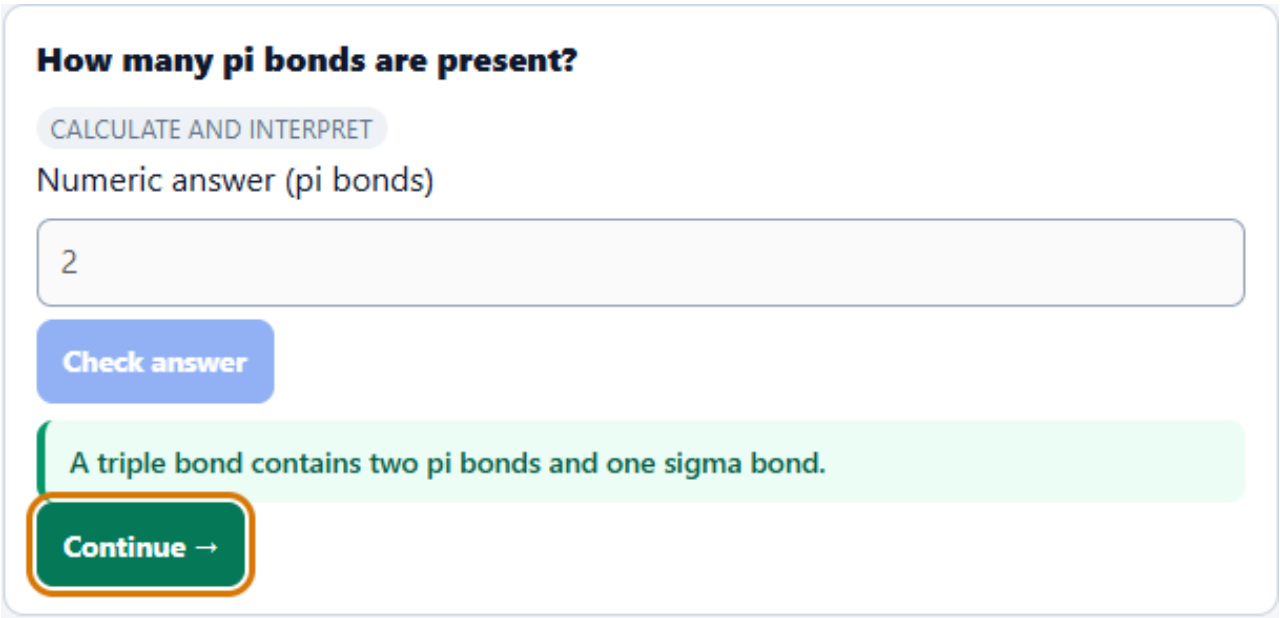
{"authorAnswer": 2, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Every bonded atom pair has one sigma bond.

Hint 2: The other two bonds are pi.

Explanation: A triple bond contains two pi bonds and one sigma bond.

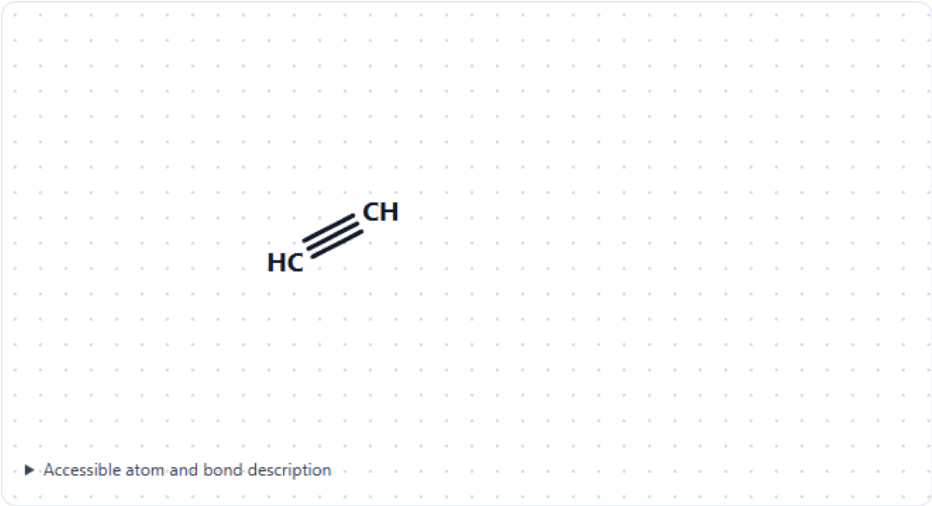
Screenshot: bond-count-stage-2-feedback



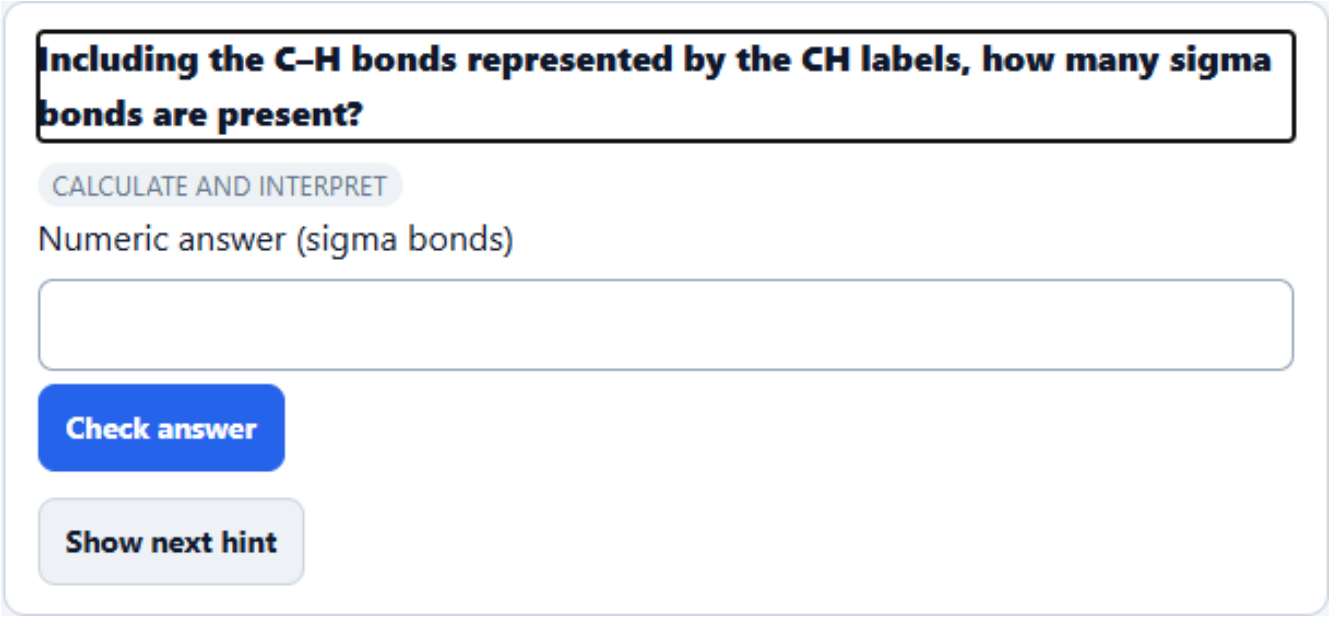
bond-count:stage-3 | Including the C–H bonds represented by the CH labels, how many sigma bonds are present?

Answer type: number | unit: sigma bonds

Screenshot: bond-count-stage-3-molecule



Screenshot: bond-count-stage-3-question



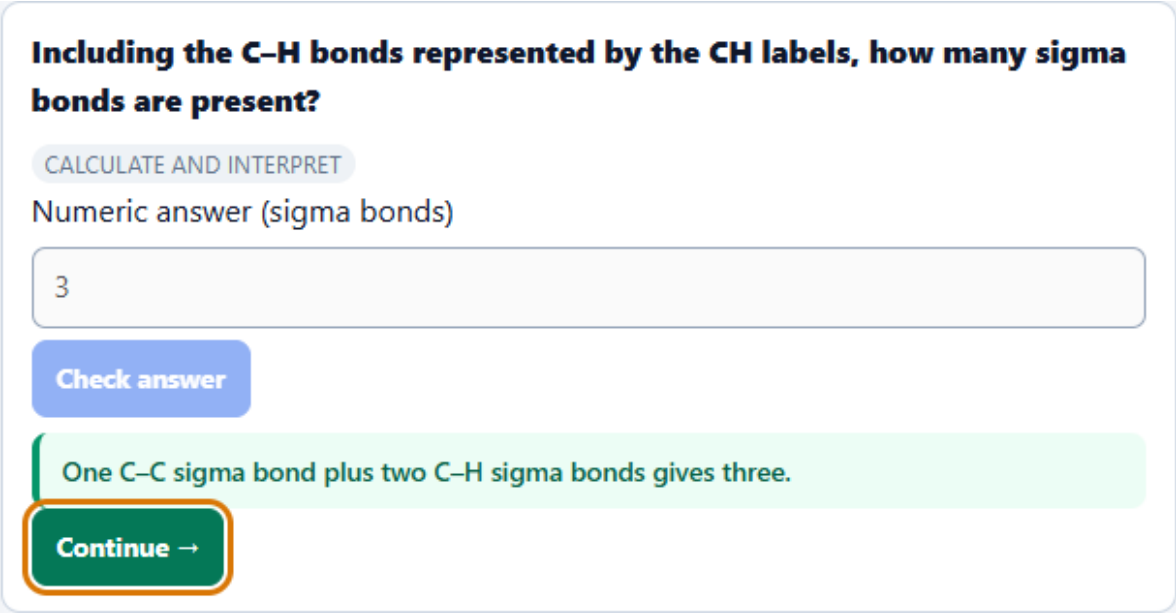
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Each CH label includes one C–H single bond not drawn as a separate line.

Hint 2: Count the C–C sigma bond once.

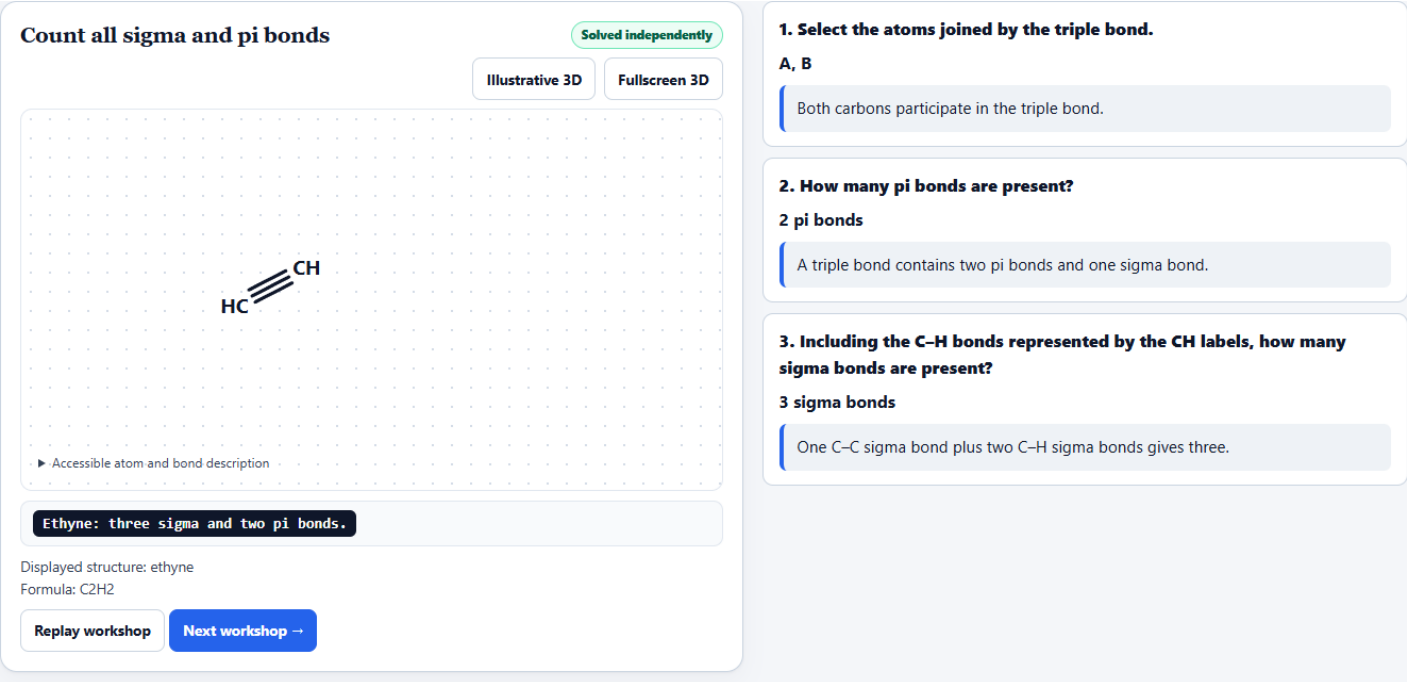
Explanation: One C–C sigma bond plus two C–H sigma bonds gives three.

Screenshot: bond-count-stage-3-feedback



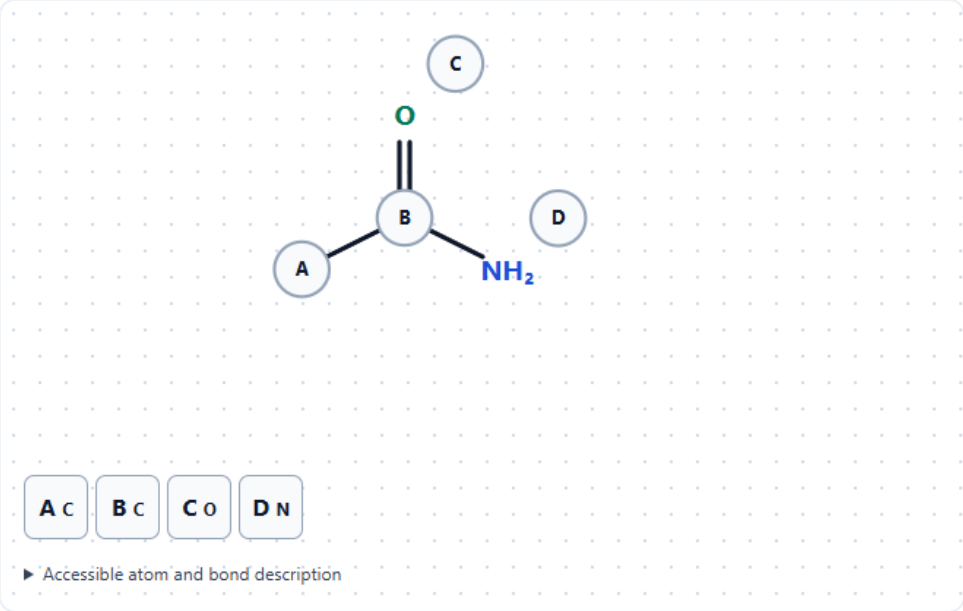
Completion summary: Ethyne: three sigma and two pi bonds.

Screenshot: bond-count-complete



amide-resonance | Use resonance to predict geometry

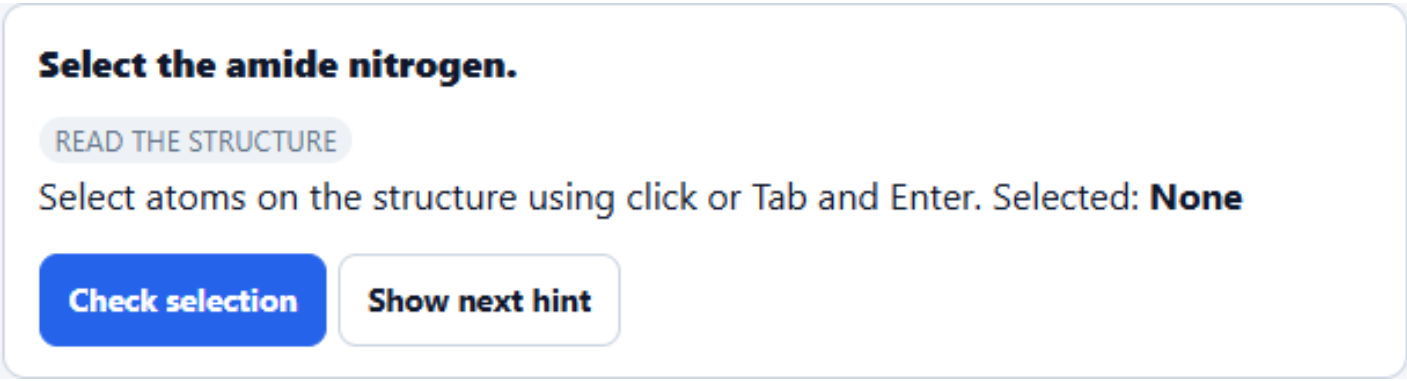
Screenshot: amide-resonance-stage-1-molecule



amide-resonance:stage-1 | Select the amide nitrogen.

Answer type: atoms

Screenshot: amide-resonance-stage-1-question



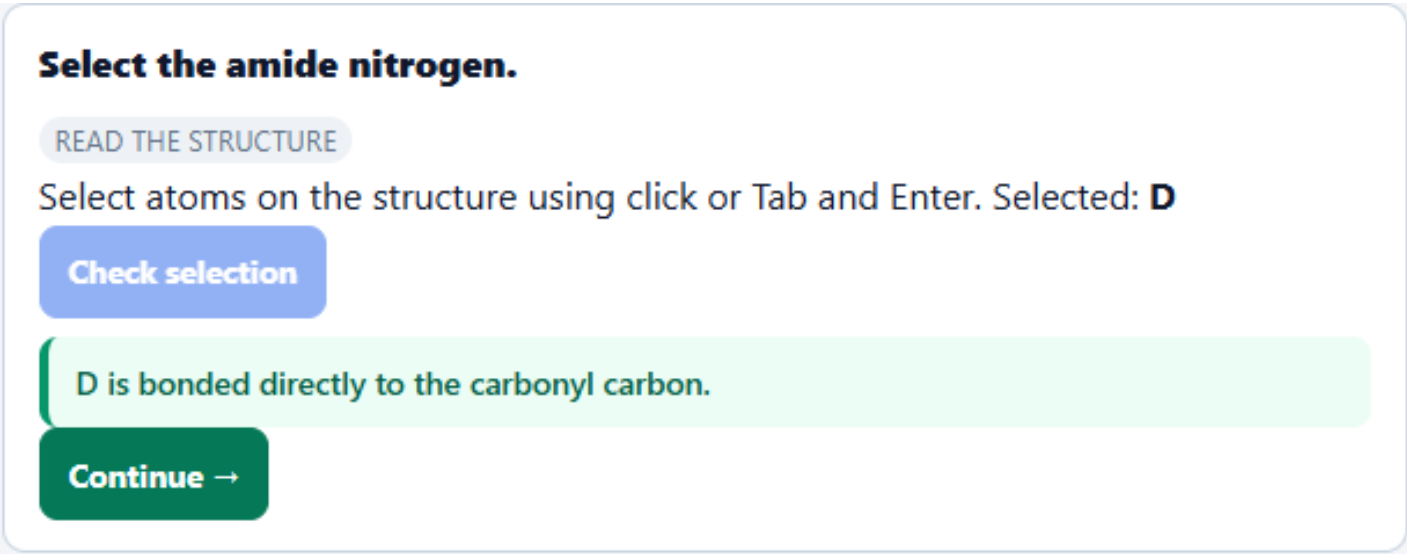
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [3]}}

Hint 1: Read the N element label.

Hint 2: The amide N neighbors the C=O carbon.

Explanation: D is bonded directly to the carbonyl carbon.

Screenshot: amide-resonance-stage-1-feedback



amide-resonance:stage-2 | Why is the amide group approximately planar?

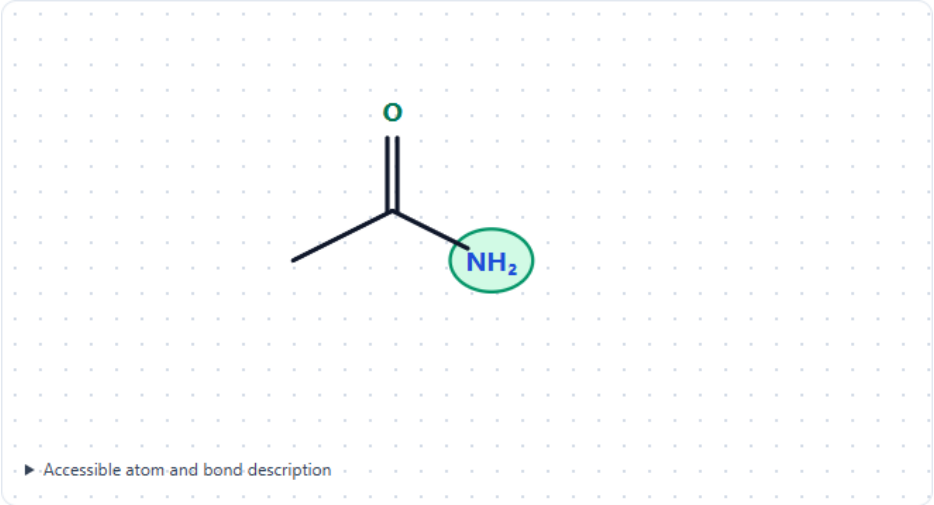
Answer type: choice

Option 0 (A): Nitrogen lone-pair conjugation with C=O

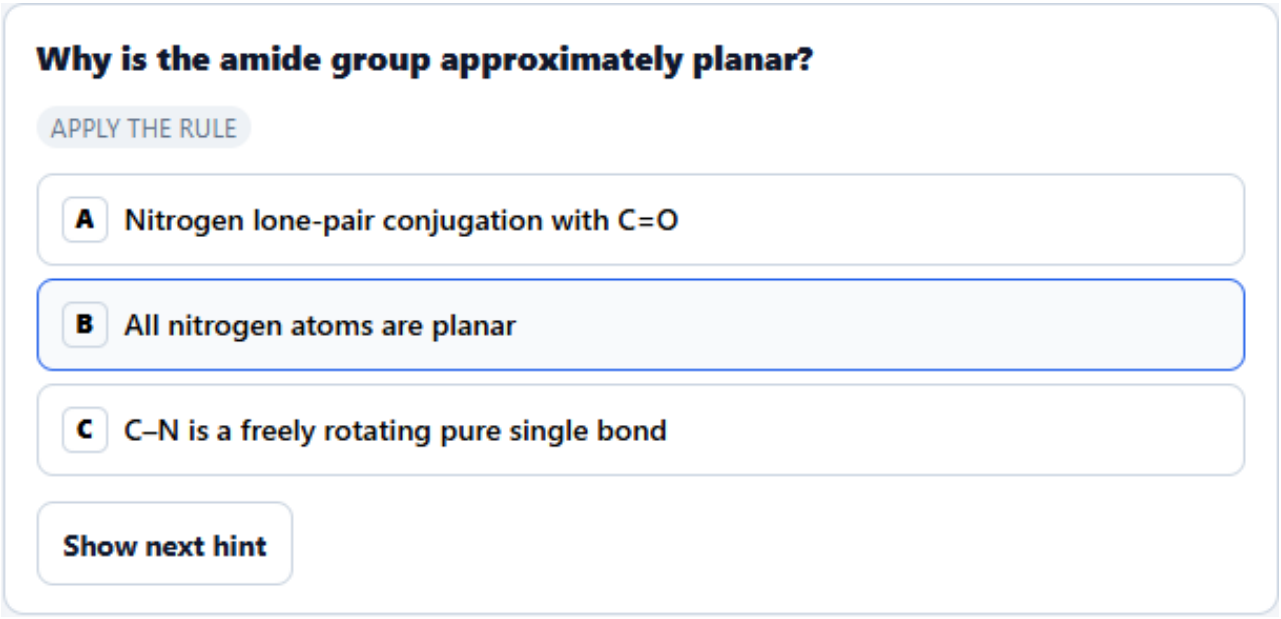
Option 1 (B): All nitrogen atoms are planar

Option 2 (C): C–N is a freely rotating pure single bond

Screenshot: amide-resonance-stage-2-molecule



Screenshot: amide-resonance-stage-2-question



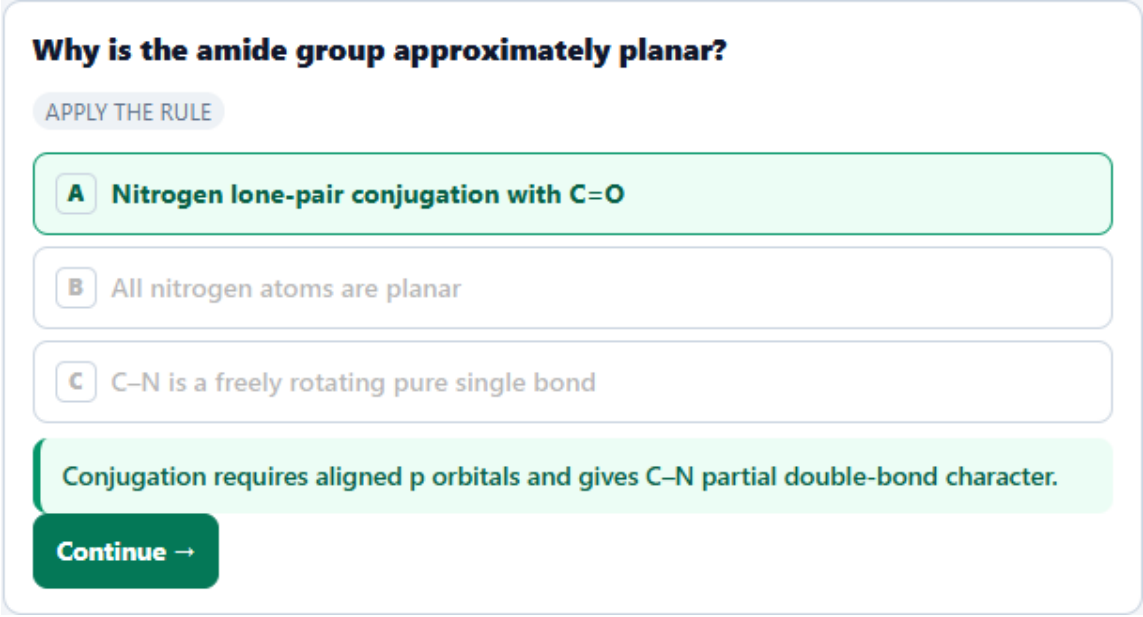
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Look for adjacent lone-pair and pi systems.

Hint 2: Delocalization requires orbital alignment.

Explanation: Conjugation requires aligned p orbitals and gives C–N partial double-bond character.

Screenshot: amide-resonance-stage-2-feedback



amide-resonance:stage-3 | For acetamide, resonance contributors differ in...

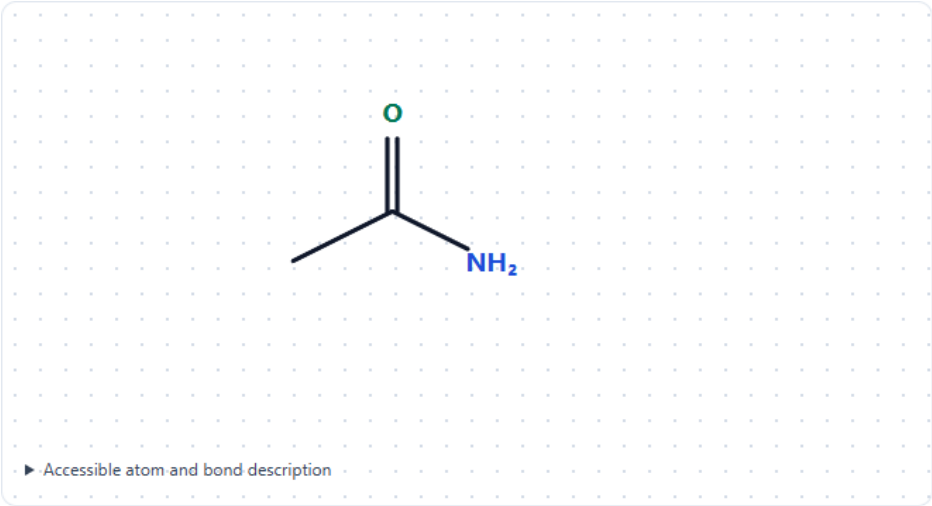
Answer type: choice

Option 0 (A): Positions of atoms

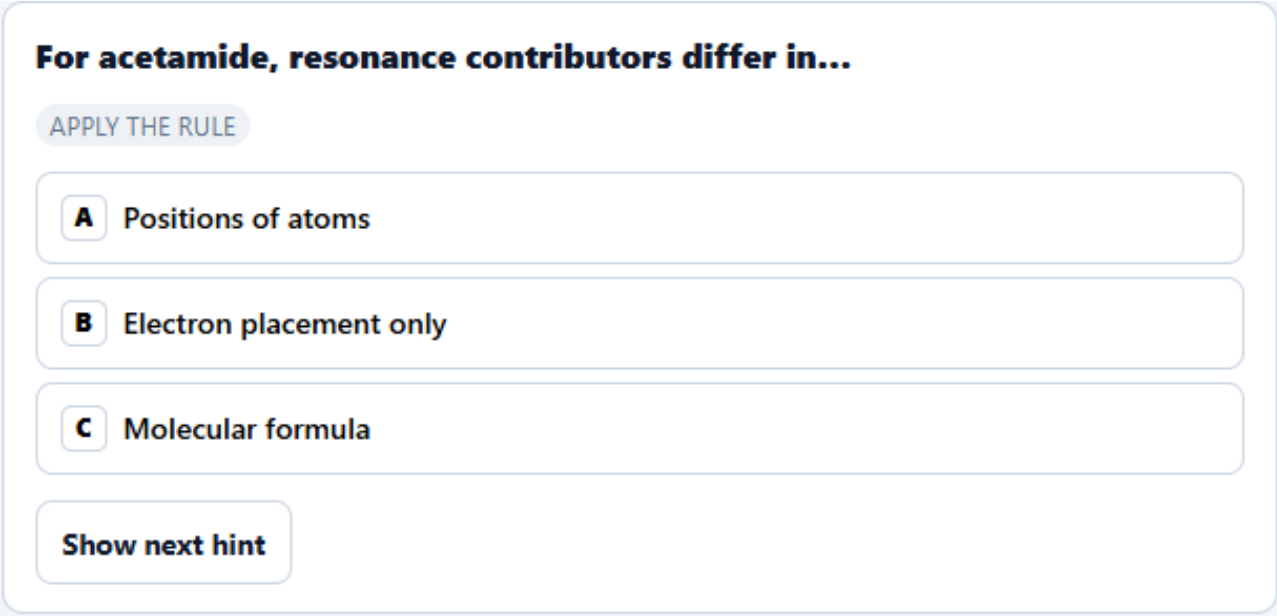
Option 1 (B): Electron placement only

Option 2 (C): Molecular formula

Screenshot: amide-resonance-stage-3-molecule



Screenshot: amide-resonance-stage-3-question



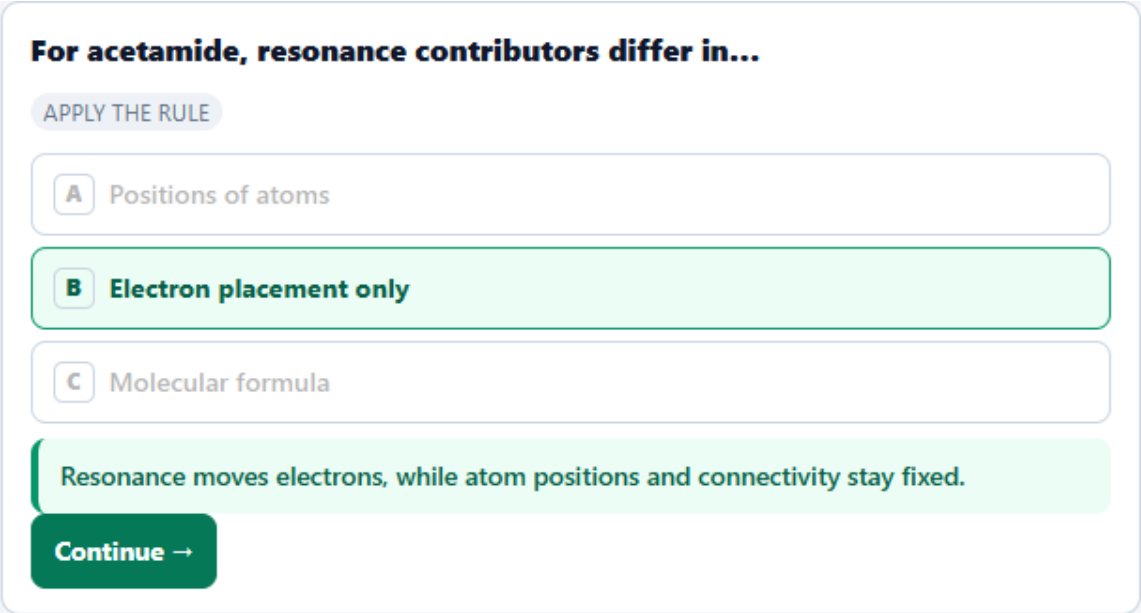
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish resonance from isomerization.

Hint 2: The sigma framework is unchanged.

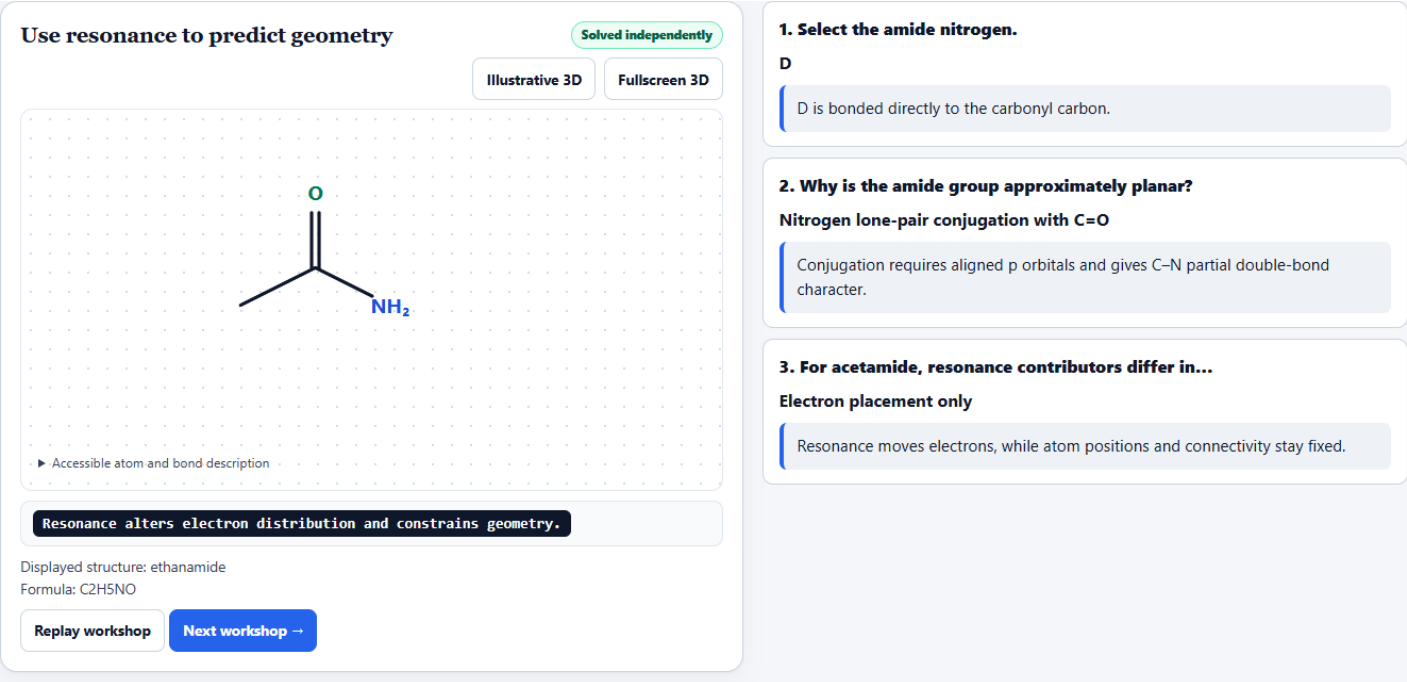
Explanation: Resonance moves electrons, while atom positions and connectivity stay fixed.

Screenshot: amide-resonance-stage-3-feedback



Completion summary: Resonance alters electron distribution and constrains geometry.

Screenshot: amide-resonance-complete



Molecular repository: author claims to verify

ethene

Author claims: ethene | formula C2H4 | canonical composition C2H4 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2]]

ethyne

Author claims: ethyne | formula C2H2 | canonical composition C2H2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 3]]

acetamide

Author claims: ethanamide | formula C2H5NO | canonical composition C2H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

acetaldehyde

Author claims: ethanal | formula C2H4O | canonical composition C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2 | canonical composition C2H3O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

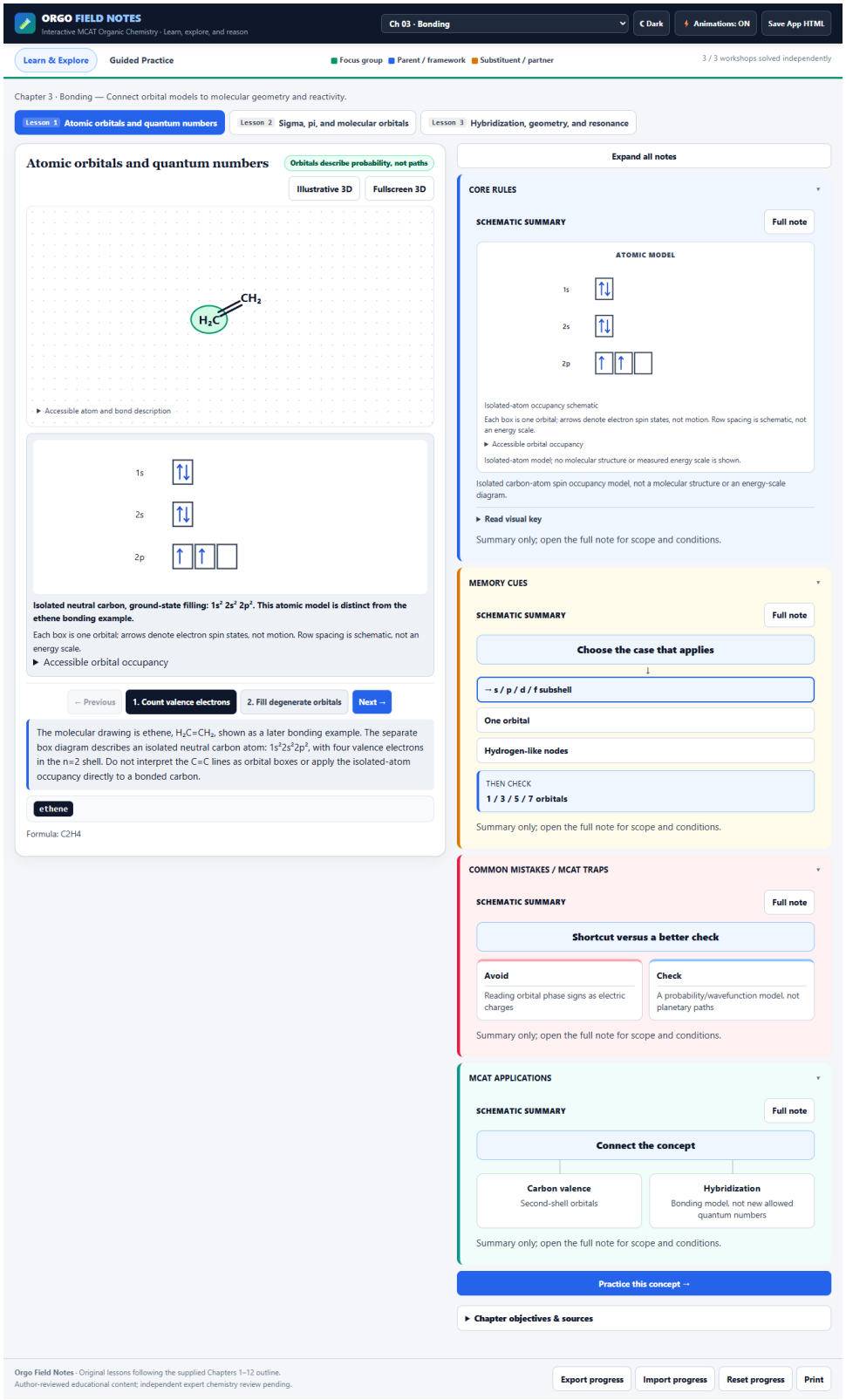
3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

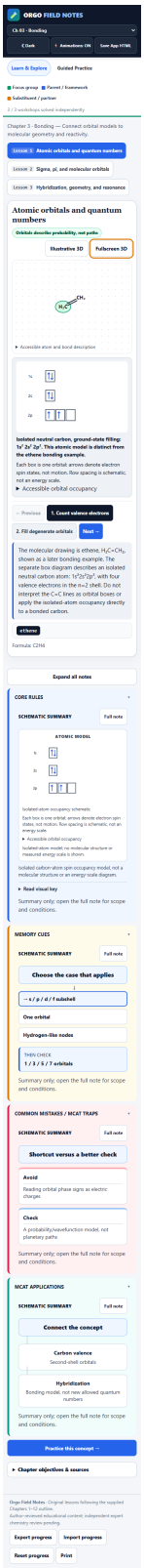
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



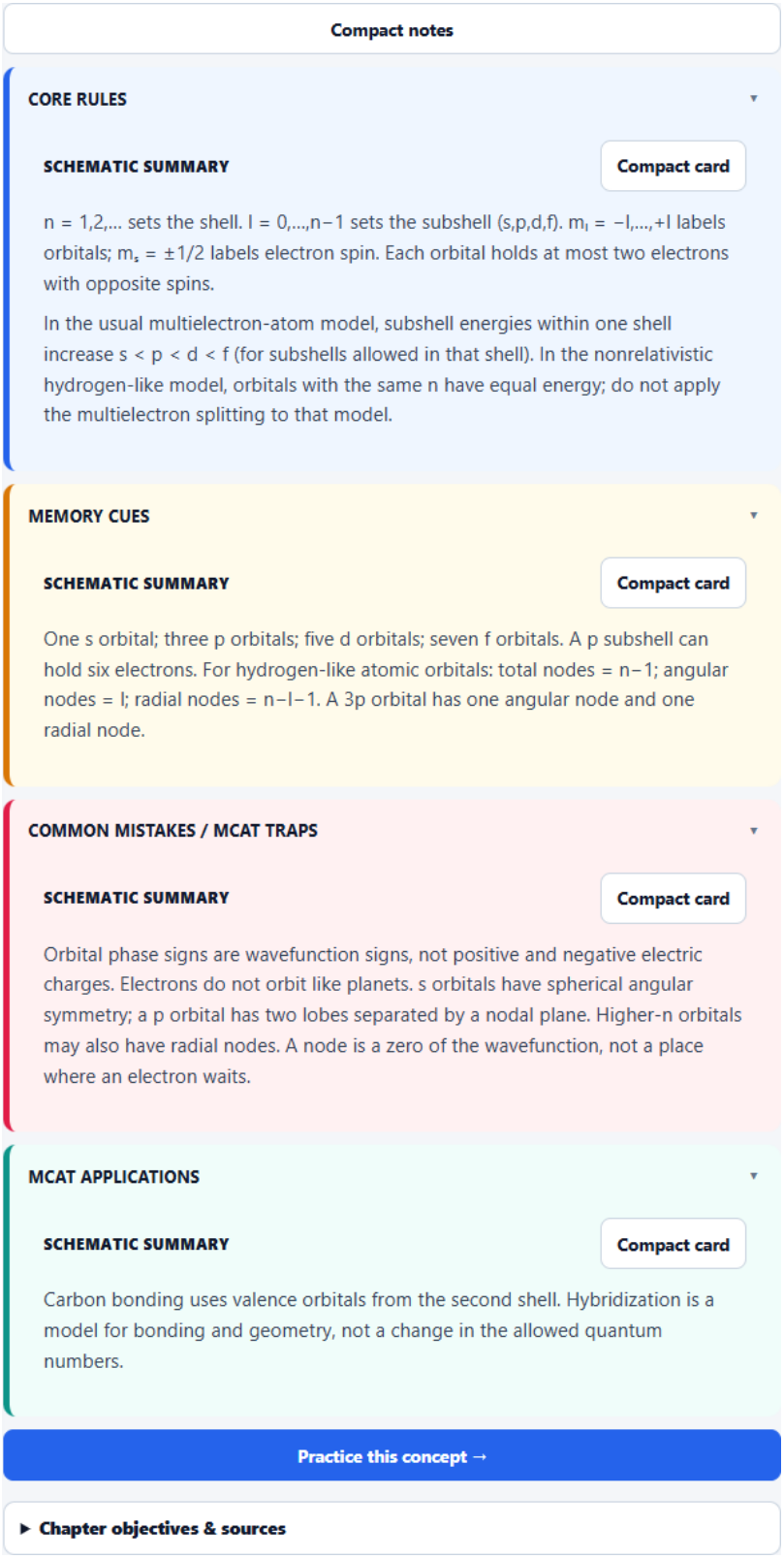
Screenshot: layout-mobile-light



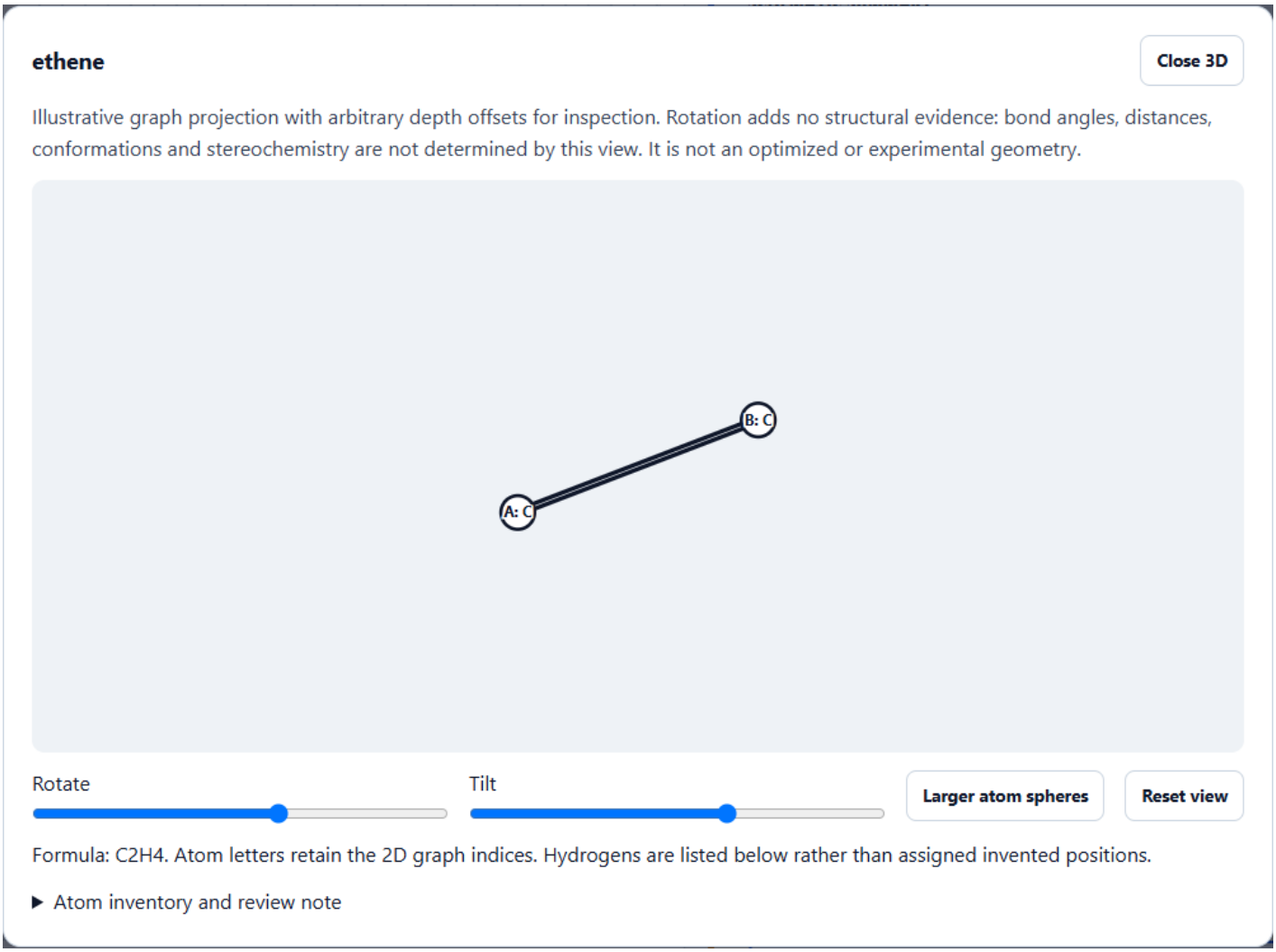
Screenshot: layout-mobile-dark



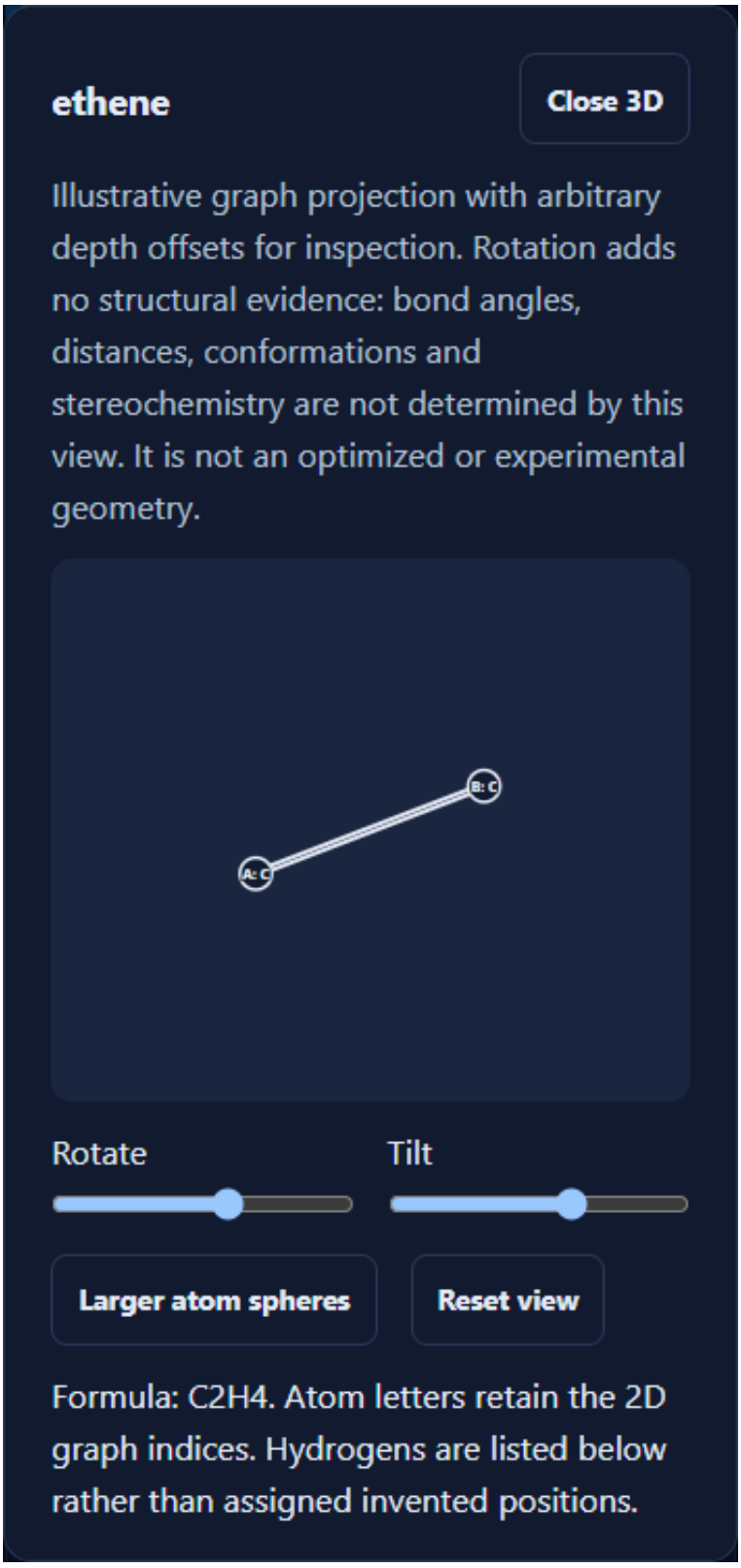
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

58 screenshots and 13 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.